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Emojis in Digital Communication: Evaluating Large Language Models for Emotional Clarity and Semantic Integrity

Hariharan Gurumurthy

Prof. Owen Conlan

August 15, 2025

A dissertation submitted to the University of Dublin, Trinity College,

In partial fulfillment of the requirements for the degree of

M.Sc. Computer Science

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Hariharan Gurumurthy

Master of Science in Computer Science
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Supervisor: Prof. Owen Conlan

Emojis have become integral to online communication, offering a compact way to express tone and emotions that pure text can sometimes obscure. Their ability to add nuance, clarify intent, and convey affect (emotions) makes them an essential part of today's digital language. This study examines whether large language models (LLMs) can use emojis to enhance the emotional clarity of text while preserving its original meaning. To test this, a four-part evaluation framework was developed, assessing output consistency, emotional appropriateness, semantic preservation, and the relationship between emotional alignment and classification accuracy.

Using a balanced, multi-emotion dataset, three LLMs were tested under four prompting strategies. Two composite measures were applied: Semantic Preservation Score (SPS) and Emoji-Emotion Congruence (EEC). The final criterion was validated through the correlation between emotional alignment and fine-grained emotion classification accuracy, with all configurations showing strong positive results ($r > 0.94$). Across models, the Emotion-Informed strategy consistently outperformed alternatives, with strategy choice proving more decisive than architecture.

These findings show that well-aligned emoji augmentation can amplify emotional signals without distorting intent, and that stronger emoji-text alignment consistently leads to higher emotion recognition accuracy. The proposed framework offers a structured, reproducible, and adaptable way to assess emotionally-aware language models, providing a basis for future research into enhancing human-computer communication through multimodal emotional cues.

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1 Introduction

The way humans communicate has undergone a significant transformation over the last two decades. Digital communication technologies like mobile messaging, social media platforms, and AI-assisted tools have proliferated into our daily lives and have redefined how we connect, express emotions, and manage interpersonal relationships with one another. Research data indicates that as of early 2024, over 60 percent of the global population uses social media (Kemp, 2024). This is a sharp increase from the earlier years, as highlighted by historical trends in social media adoption (Ortiz-Ospina, 2019). Confined face-to-face interaction or traditional written correspondence are no longer the case, and text-based, image-enhanced digital modalities are the new trends.

While digital communication offers convenience and immediate delivery, it also presents new challenges like emotional clarity. The absence of tone, facial expression, and gesture creates a situation where the interpretation of emotional intent becomes harder, increasing the risk of miscommunication and the related friction (Walther, 2011; Kruger, Epley, Parker, & Ng, 2005). Virtual empathy-based research (Čekić, 2025) further suggests that these digital platforms not only mediate such interactions but also influence how their users regulate emotion and perceive social connections. One consequence of this is what is termed “emotional ambiguity crisis” – difficulty in conveying and interpreting emotion through brief text-based exchanges, which is mentioned in detail in the upcoming section.

1.1 The Emotional Ambiguity Crisis in Digital Communication

Consider receiving a text message from a colleague that reads,

“Nice job on the presentation.”

Without any vocal inflection or facial expressions, the message becomes emotionally ambiguous. Is it a sincere praise, a subtle sarcasm, or restrained disappointment? The same five words could express a genuine approval, a gentle teasing, or an even passive criticism, all depending entirely on the sender’s unspoken emotional state.

The above scenario is a classic example of “*emotional ambiguity crisis*” - a systemic issue in contemporary communication where users must navigate complex emotional meaning with limited cues (Walther, 2011; Byron, 2008) .

One can argue that this is not just a problem of the digital era, and that is true. The difficulty of interpreting emotional intent has always existed in written text communications. Traditional methods like letters and emails, as well, did not possess the necessary paralinguistic cues like tone, etc., making the reader assume the affective meaning. Furthermore, people also often misjudge how well others understand their tone in text-based communication (Kruger, Epley, Parker, & Ng, 2005).

However, with digital communication, this ambiguity is both amplified and highly pervasive. Unlike formal writing, digital exchanges are rapid, informal, and often stripped of context, increasing the risk of emotional misinterpretation (Walther, 2011; Baron, 2008). The speed at which digital messaging happens effectively creates an environment where it simulates an in-person conversation, without the facial and vocal cues. Studies also show that the lack of these paralinguistic signals imposes a greater cognitive load, increasing the chances of misreading the emotional intent and thus affecting the relative outcome (Derks, Fischer, & Bos, 2008). With billions of such interactions happening daily, emotional ambiguity is no longer an individual’s crisis but now a societal concern for digitally mediated communication (Callais, 2020).

1.2 Emojis as Linguistic Bridges: Promise and Complexity

In order to overcome the emotional ambiguity in digital communication, users have been increasingly relying on emojis to compensate for the absence of non-verbal cues in text-based interactions. Evolving from early emoticons (e.g., :-) or ;-) , emojis (😊, 😂, 🙄, etc.) - a bunch of visual symbols are now recognized as a form of digital paralanguage- work to supplement text with affective or contextual cues (Gawne & McCulloch, 2019). Introduced in Japan in 1999, emojis have become a global phenomenon, with over 3600 standardized symbols used daily across billions of messages (Evans, 2017).

Although they may appear as decorative flourishes, emojis function as affective signals that significantly shape message interpretation. For example, adding a ‘😊’ to “Nice job on the presentation” might signal warmth, 🙄 could suggest sarcasm, and 🤔 might imply

polite discomfort. A casual scroll through social media can reveal just how slippery these meanings can be. Take the case of 😬 emoji, for example. It might express nervousness, shared embarrassment, polite awkwardness, or even low-key amusement; all depending entirely on the message, the sender, and the vibe. Used with *"I told my boss 'love you' by mistake"*, 😬 reflects social panic and embarrassment; whereas used in *"I ate 32 nuggets in one sitting"*, it's a proud but cringing celebration. Even the humble 👍, a universal sign of agreement, can feel dismissive, sarcastic, or cold in certain contexts.

Emojis, in short, are not simple but are as complex as emotions themselves. They blend social cues, tone, and intent, often expressing multiple emotions at once rather than mapping neatly onto a single feeling (Kerslake & Wegerif, 2017). This emotional richness, however, comes at a cost: emojis are not universally understood. Their meaning often depends not just on the context, but also on the cultural norms, platform-specific rendering, and generational usage. For example, the 🙏 emoji may be understood as gratitude / pray in certain cultural contexts but as a high-five in others; the 😂 emoji, once widely associated with laughter, is increasingly used by younger users to signal irony or ridicule. Such shifts are amplified furthermore by design inconsistencies across platforms (the way they look across platforms like Apple, Android, and Web are different), thus altering their perceived intent. Studies have found that emoji misinterpretation rates can reach as high as 75% in cross-platform communication, particularly in emotionally charged or ambiguous contexts (Miller, et al., 2016; Tigwell & Flatla, 2016).

These interpretive challenges extend beyond casual misunderstanding. Ambiguous emoji use has been cited in legal proceedings (Kirley & McMahon, 2017), and some symbols have been repurposed as coded communication in illicit digital spaces (Alhendi, Baniamer, Alsalamat, Salameh, & Alanazi, 2024). Such variability makes emotional intent difficult to infer, especially for computational systems. As machines increasingly augment or generate human messaging, the ability to interpret emoji meaning with contextual precision becomes not only a technical challenge, but a communicative imperative.

1.3 LLMs and the Emotional Intelligence Gap

Digital platforms today are dominated by emotionally saturated, short-length textual contents like tweets, captions, and status updates. These bite-sized messages often rely on condensed emotional cues, including emojis, to convey tone, stance, or affect.

However, their brevity makes them especially vulnerable to misinterpretation, particularly when nonverbal or paralinguistic cues are absent.

In the current emotionally charged digital landscape, Large Language Models (LLMs) have already emerged as influential co-authors. Deeply embedded into the content creation pipelines, LLMs nowadays produce everything from influencer-like captions to political rhetoric and viral posts, all skillfully replicating emotional tone and stylistic subtleties (Coppolillo, Cinus, Minici, Bonchi, & Monco, 2024). Beyond linguistic coherence, these LLMs are also systematically trained to calibrate emotional expression following platform-specific engagement patterns, thereby optimizing the content for visibility, interaction, and resonance with target audiences (Radivojevic, Chou, Badillo-Urquiola, & Brenner, 2024)

While the expressive capabilities of LLMs offer notable communicative advantages, they also introduce a few significant areas of concern. Research demonstrates that social media platforms disproportionately reward emotionally charged content, particularly the ones that exhibit outrage, moral grandstanding, or affective polarization. LLMs are prone to emulate these patterns and amplify them through algorithmic reproduction (Milli, Carroll, Pandey, Wang, & Dragan, 2023). This is further intensified by the integration of advanced personalization tools, enabling users to craft emotionally impactful and compelling contents that simulate authenticity in the absence of genuine affective intention (Zhao, et al., 2025).

The complexity of computational affect becomes even more pronounced in the context of emoji use. Unlike human content creators, who draw upon rich relational, cultural, and contextual cues in their selection of emojis, LLMs often operate via reductive sentiment-based heuristics. This can lead to misalignments between the intended affective message and its representation, potentially resulting in ambiguity or distortion. As LLMs become more deeply embedded in everyday communication, it is crucial to assess whether their use of emotional cues, including emojis, preserves the emotional clarity and aligns with human communicative norms.

1.4 Research Objectives, Research Question, and Hypotheses

This study investigates whether Large Language Model (LLM) generated emoji augmentation can enhance emotional clarity while preserving contextual meaning in digital communication. The work is guided by a structured evaluation framework that defines and measures the communicative effectiveness of emoji augmentation; specifically, how well LLM-generated emojis support emotionally transparent and semantically faithful messaging.

1.4.1 Research Gap

Prior research in sentiment analysis, emotion analysis, emoji prediction, and emoji usage patterns has largely treated text and emojis as separate computational tasks. Moreover, there is limited empirical evidence on how large language models (LLMs) generate and integrate emojis in ways that affect both emotional interpretation and semantic preservation.

1.4.2 Research Objectives

This study addresses the above gap by developing a replicable evaluation framework grounded in affective communication theory and by empirically analyzing LLM-driven emoji usage in diverse emotional contexts. The objectives are:

O-1: Identify and analyze the literature for gaps in computational affective communication, with a focus on emoji usage, LLM-driven augmentation, and existing evaluation frameworks.

O-2: Design and implement a replicable evaluation framework that assesses both emotional congruence and semantic preservation in LLM-generated emoji augmentation.

O-3: Identify, justify, and evaluate a balanced, multi-emotion dataset suitable for testing LLM-driven emoji augmentation, ensuring coverage of primary, secondary, and tertiary emotional dyads.

O-4: Compare and analyze the performance of multiple LLM architectures and prompting strategies using the proposed framework, with metrics covering EEC, SPS, including measures of consistency, appropriateness, preservation, and intensity

O-5: Examine performance variation across basic, secondary, and tertiary emotional dyads, identifying patterns and challenges in handling increasing emotional complexity.

O-6 Understand the relationship between emotion–emoji alignment and downstream fine-grained emotion classification accuracy, validating whether stronger congruence correlates with improved recognition performance.

O-7 Identify practical implications for responsible emotional AI development, including cultural sensitivity, user–AI collaboration, and ethical considerations for deployment.

1.4.3 Core Research Question and Hypotheses

This study addresses the following central question:

Can an LLM effectively augment plain text with emojis?

Effectiveness, in this context, is defined as successful emoji augmentation, operationalized through four empirically grounded evaluation dimensions and the corresponding hypotheses.

1. Consistency: Whether the LLM assigns similar emojis to emotionally or structurally similar texts, reflecting reliable behavior.
 - **H1: Consistency Hypothesis:** LLMs will demonstrate **reliable emoji placement patterns** when processing semantically similar texts with identical emotional labels.
2. Appropriateness: Whether the generated emojis correspond to the intended or labeled emotional tone of the message.
 - **H2: Appropriateness Hypothesis:** LLM-generated emojis will demonstrate high emotional congruence with the original text’s emotional content.
3. Preservation: Whether emoji augmentation maintains the original semantic content and communicative purpose of the text.
 - **H3: Preservation Hypothesis:** Emoji augmentation will maintain semantic integrity without distorting the original message meaning.

4. Emotional Intensity Association: Does stronger emotion–emoji alignment correspond to higher emotion classification performance?
 - **H4: Intensity Hypothesis**: Texts where emojis are more closely aligned with the intended emotional tone will achieve better downstream emotion classification accuracy.

Together, these objectives and hypotheses establish a clear framework for connecting computational performance with meaningful human communication outcomes.

1.5 Key Contributions

This study makes several contributions across theoretical, methodological, and practical domains. Theoretically, it introduces a communicative lens on emoji augmentation by framing emojis as digital paralinguage, expanding existing understandings of emotional expression in text. It also develops a clear, multidimensional framework, focused on Consistency, Appropriateness, Preservation, and Interpretive Impact to evaluate the effectiveness of LLM-generated emoji usage.

Methodologically, the study proposes a replicable evaluation pipeline that integrates emotion-labeled datasets, semantic similarity metrics, and human judgment tasks. This mixed approach captures both technical output and its perceived emotional clarity from a user-centered perspective.

Practically, the findings offer concrete insights for improving LLM-driven communication tools and highlight key risks such as emotional mismatch or semantic drift (i.e., emojis that send mixed signals or change the meaning of the message) in current emoji augmentation strategies. Overall, the work advances a more nuanced understanding of how language models influence emotional communication in digital contexts.

Together, these foster a rich, more practical understanding of how language models can shape our emotional expression in digital communication spaces both positively and negatively.

1.6 Thesis Structure

This thesis charts a logical progression from conceptual groundwork through methodological design and empirical investigation, ultimately leading to practical insights on how Large Language Models (LLMs) augment digital communication with emojis. Each chapter aligns with one or more of the study's research objectives (O-1 to O-7), ensuring a direct link between the research aims and their fulfilment.

- **Chapter 2** surveys the theoretical and empirical foundations of emotion theory, digital expression, emoji research, and LLM-based content generation. It identifies key methodological gaps and establishes the conceptual base for the proposed evaluation framework.
- **Chapter 3** introduces the dual-metric evaluation framework: Semantic Preservation Score (SPS) and Emoji–Emotion Congruence (EEC), supported by a Quality Assurance (QA) system. It operationalizes the four evaluation dimensions (Consistency, Appropriateness, Preservation, Interpretive Impact) and details the theoretical rationale, computational pipeline, and metric integration.
- **Chapter 4** outlines the experimental design, covering dataset selection, preparation, and justification, as well as model choice, prompt engineering strategies, and configuration controls. It explains the evaluation pipeline used to measure performance across diverse LLM architectures.
- **Chapter 5** presents and analyses the empirical results, comparing model outputs and interpretive alignment. It explores performance trends, context-driven variations, and divergence patterns, validating the research hypotheses through both quantitative and qualitative evidence.
- **Chapter 6** reflects on the broader significance of the key findings, discussing their implications for affective computing and AI-mediated communication. It also addresses the study's limitations and highlights future research directions.
- **Chapter 7** concludes the thesis by synthesizing the core contributions, revisiting the central research question, and underscoring the importance of designing emotionally attuned AI systems that can support more expressive and context-aware digital interactions.

2 Literature Review: Theoretical Foundations and Related Work

This chapter establishes the basis for the study by surveying key emotion theories, the rise of emojis as emotional cues in digital communication, and advances in emotion and sentiment analysis systems. It highlights the persistent challenge of aligning LLM-generated emoji usage with genuine emotional intent, underscoring the need for a context-aware framework to evaluate the effectiveness of emoji augmentation in AI-mediated environments.

2.1 Emotion Theory and Digital Expression

The theoretical foundation for understanding emotions in digital communication draws from established psychological frameworks adapted for computational contexts. Plutchik's psycho-evolutionary model (Plutchik, 1980) continues to be one of the most influential frameworks in the field of emotion science and computational affect analysis. Based on the notion that emotions serve adaptive evolutionary functions, the model introduces eight basic emotions: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation, arranged in four bipolar pairs. These are visualized in his iconic "wheel of emotions" (Figure 2.1), which organizes feelings not only by opposition but also by intensity (e.g., joy <-> ecstasy <-> serenity).

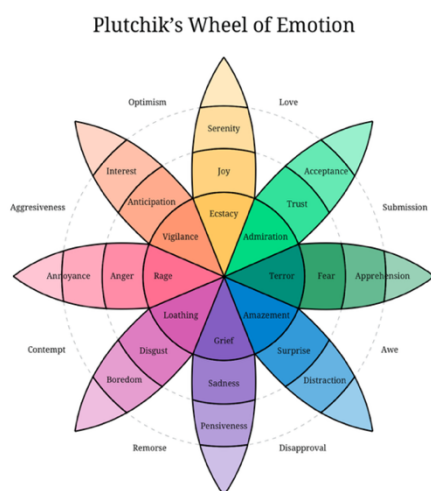


Figure 2.1 Plutchik's wheel of Emotions

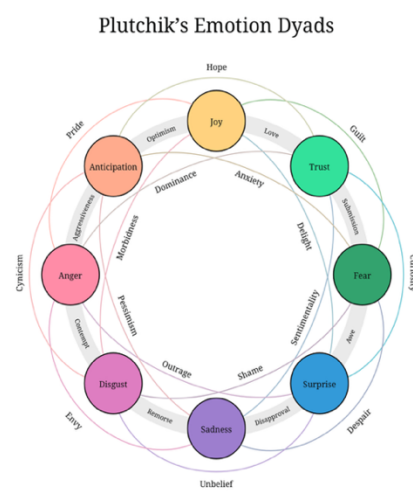


Figure 2.2 Plutchik's Emotion Dyads

A unique strength of this model lies in its notion of emotion dyads (refer to Figure 2.2), which describes how adjacent emotions can come together to form complex emotional states. For instance, joy and trust may blend to form love (a primary dyad), while more distant pairs give rise to secondary and tertiary dyads such as guilt (joy + fear) or shame (fear + disgust). Because of its hierarchical and combinatorial structure, this model is particularly adaptable in digital contexts where emotional expression is often compact, multimodal, and stylistically varied. In computational linguistics and AI applications, particularly in systems that augment language with visual or symbolic cues like emojis, Plutchik's framework enables more nuanced tagging, synthesis, and interpretation of affect. The influence of this model is evident across datasets, emotion classifiers, and emoji-emotion mappings, making it a foundational reference point for evaluating emotional coherence and complexity in AI-mediated communication.

Russell's dimensional approach provides a complementary framework for mapping emotions within a continuous two-dimensional space defined by valence (pleasure–displeasure) and arousal (activation–deactivation). In it (Russell, 1980), Russell demonstrated that emotions can be positioned in a circular arrangement around these two bipolar axes, allowing for systematic quantification and comparison of emotional states (Figure 2.3).

Valence in the context of emotion science refers to the intrinsic positivity or negativity of an emotion, ranging from pleasant feelings like joy and love to unpleasant ones like anger or sadness. Similarly, Arousal measures the level of activation or energy associated with the state, from high (excitement, fear) to low (calmness, boredom). This allows for a nuanced representation of emotional experience, capturing distinctions between emotions with similar valence but different arousal (e.g., excitement vs. contentment). As such, this valence-arousal model has become foundational in many sentiment and emotion analysis systems, offering a robust mathematical basis for representing and analyzing emotional content in computational contexts.

Ekman's basic emotion theory establishes a set of universal emotional categories through extensive cross-cultural research. It (Ekman, 1992) identifies six fundamental emotions (anger, disgust, fear, happiness, sadness, and surprise), each linked to distinctive facial expressions recognized across cultures. While foundational in early emotion recognition systems, this model faces limitations with respect to capturing the

full complexity of emotions in digital environments. In emoji-rich contexts, where emotions are conveyed through symbolic and often culturally nuanced cues, such universal categories may fall short in accounting for variation in interpretation.

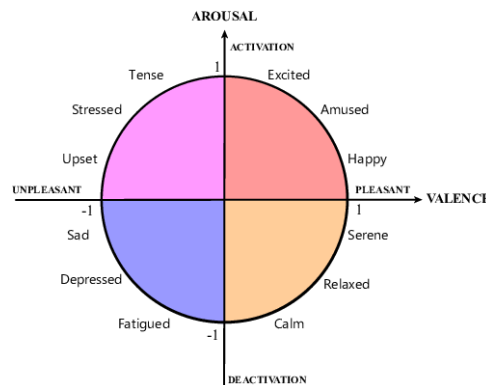


Figure 2.3 Russell's two-dimensional emotion model

The shift from a psychology-based understanding of emotions to digital forms of expression, such as online chats, social media, and emojis, requires hybrid approaches that combine psychological theory with computational methods. Frameworks for digital emotional expression must address the symbolic role of emojis as emotional signifiers, cultural variations in interpretation, and emerging modalities unique to online communication.

2.2 The Evolution of Emoji Research

Research on emojis has progressed significantly from basic cataloging of symbols to examining their complex role in the expression and interpretation of emotions in digital communication. Initially, emojis were regarded as decorative or optional elements within digital exchanges; however, they are now recognized as essential components for conveying nuance, humor, and affect in text.

A seminal study (Novak, Smailovic, Sluban, & Mozetic, 2015) marked a pivotal advancement with the development of the initial large-scale Emoji Sentiment Ranking. Their analysis encompassed over 1.6 million tweets across 13 languages, systematically documenting the emotional polarities associated with 751 frequently used emojis and demonstrating that, despite varied user contexts, several emotional associations exhibit stability. While positive sentiment was prevalent, the study also highlighted significant

contextual influences on interpretation, indicating that the emotional connotation of emojis is context dependent.

Building upon this foundation, a novel computational approach to emoji semantics was introduced through vector embeddings. Using the Emoji2Vec model (Eisner, Rocktäschel, Augenstein, Bošnjak, & Riedel, 2016), vector representations were constructed for over 1,600 emojis, derived from Unicode descriptions rather than empirical usage frequency. This methodology facilitated the treatment of emojis as semantically quantifiable entities, thus enabling more refined semantic and similarity analyses between emojis and lexical items.

Subsequently, (Wijeratne, Balasuriya, Sheth, & Doran, 2017) developed EmojiNet, a comprehensive machine-readable inventory associating approximately 2,400 emojis with over 12,000 potential meanings, incorporating definitions from lexical resources such as BabelNet. This work bridged symbolic emoji representations with their real-world linguistic interpretations, providing resources applicable to both academic inquiry and practical applications.

Collectively, these developments illustrate a clear trajectory. Research on emojis has moved beyond mere cataloging and labeling, and now increasingly depends on analyzing the interplay between emojis and textual content. This shift has revealed layered meanings and affective nuances in digital communication. It furnishes the theoretical basis for contemporary studies, such as the present research, to develop evaluation frameworks that consider emoji use within the full scope of digital communicative contexts rather than treating emojis as isolated symbols.

2.3 Sentiment vs. Emotion Analysis Systems

The difference between sentiment and emotion analysis is imperative for this research work as computational models evolve beyond basic polarity classification. While sentiment analysis focuses primarily on the overall tone of a message: typically positive, negative, or neutral, the emotion analysis targets the identification of specific emotional states and their relative intensities. Liu's foundational taxonomy (Liu B. , 2012) clearly articulates these differences, marking an essential theoretical turning point. This

distinction becomes especially salient in emoji-rich contexts, where multiple emotions may coexist, and where emotional intensity and nuance are central to meaning-making.

2.3.1 Text-Based Emotion Detection Systems

One of the most widely adopted systems for fine-grained emotion recognition is GoEmotions (Demszky, et al., 2020). Comprising about 58,000 manually annotated Reddit comments classified across 27 distinct emotion categories plus a neutral label, GoEmotions has become a benchmark for evaluating emotion classification in English-based online content. The model's strong performance, achieving an average macro F1-score of 0.46 using a BERT-based architecture, demonstrates its utility for text-based emotion detection at scale. However, a significant limitation lies in its treatment of multimodal context: any emoji present in the original comments is removed during preprocessing, thereby excluding symbolic affective cues that often shape or intensify emotional interpretation. As a result, GoEmotions provides no mechanism for analyzing the alignment or interaction between textual and emoji-based emotional signals.

Similarly, the EMO-KNOW dataset (Nguyen, Samaradivakara, Sasikumar, Gupta, & Nanayakkara, 2023) offers another significant contribution through its annotation of emotion-cause pairs, enabling causal analysis of affective expression in tweets. With over 700,000 annotated instances covering 48 emotion types, EMO-KNOW advances methodological depth by capturing emotional triggers within short-form content. Yet, like GoEmotions, it systematically removes emojis from raw data during preprocessing, treating them as extraneous noise. This kind of exclusion is prevalent across many emotion datasets, which effectively renders the system blind to multimodal expression, discounting the prominent role emojis often play in conveying or amplifying emotional meaning in digital interactions.

2.3.2 Emoji-Focused Analysis Systems

On the other end of the spectrum, some systems emphasize emoji as emotionally meaningful units. DeepMoji (Felbo, Mislove, Søgaard, Rahwan, & Lehmann, 2017) was trained on a vast dataset of 1.246 billion tweets, where emojis were used as *weak supervision signals*. Here, emojis were not explicitly assigned by annotators as emotion tags but were assumed to reflect the general sentiment or emotional tone of the tweet. For example, if a tweet included a 😞 emoji, the model treated that as a signal that the

text expressed sadness, even though that might not always be strictly accurate. Despite their imprecision, these emoji signals allowed the model to learn emotional associations from an enormous volume of real-world data, capturing subtle patterns in how people express affect in text.

What makes it influential is its ability to transfer the emotional understanding it gained from emoji-rich data onto other emotion-related tasks (also known as transfer learning). However, its approach remains fundamentally predictive. It is designed to infer which emoji best fits a given piece of text. Still, it does not evaluate whether the selected emoji, once inserted back into the message, improves emotional clarity, maintains semantic intent, or aids human understanding. There is no mechanism within the model to assess how well the predicted emoji aligns with the communicative purpose of the original text. This limitation is particularly pertinent as we progress towards generative systems like LLMs that not only predict but also generate augmented content. In these cases, evaluating the interpretive impact of emoji augmentation, rather than just its statistical accuracy, becomes increasingly important.

EmoTag1200 (Shoeb & de Melo, 2020) presents the most extensive human-labeled dataset mapping 1,200 emojis to Plutchik's eight primary emotions with the help of intensity scores. This resource offers a rich basis for comparing emoji semantics against a well-theorized emotional framework. Yet, despite its theoretical rigor, EmoTag1200 focuses solely on the emoji side of the equation; it does not evaluate how well these intensities align with the emotions expressed in real-world textual usage. The dataset provides valuable information about how individual emojis are perceived in isolation, but remains agnostic about how they interact with language in actual communicative contexts.

2.3.3 Limitations in Current Approaches

Despite the progress made, current systems exhibit systematic limitations that restrain their utility for context-sensitive emoji evaluation. Text-only models like GoEmotions and EMO-KNOW neglect the affective signals encoded in emoji use, resulting in incomplete representations of emotional intent. Emoji-only systems such as EmoTag1200 and the Emoji Sentiment Ranking (Kralj Novak et al., 2015) work in isolation from language, failing to account for how emoji meanings are shaped or modified by adjacent text. Predictive

frameworks like DeepMoji focus on generation but lack evaluation mechanisms for emotional appropriateness after augmentation. Finally, embedding-based models (e.g., Emoji2Vec) treat emojis as semantic units but offer no protocols for assessing congruence between emoji and surrounding text in real-world contexts.

<i>System</i>	<i>Authors</i>	<i>Dataset Size</i>	<i>Primary Function</i>	<i>Key Strength</i>	<i>Critical Limitation</i>
<i>GoEmotions</i>	(Demszky, et al., 2020)	58,009 comments	Fine-grained text emotion detection (27 categories)	High-quality emotion taxonomy with validated annotations	Completely ignores emoji-text alignment; processes text as if emojis absent
<i>DeepMoji</i>	(Felbo, Mislove, Søgaard, Rahwan, & Lehmann, 2017)	1.246 billion tweets	Emoji prediction as distant supervision	Excellent transfer learning across 8 benchmarks	No assessment of predicted emoji appropriateness in original context
<i>EMO-KNOW</i>	(Nguyen, Samaradivakara, Sasikumar, Gupta, & Nanayakkara, 2023)	700,000+ tweets	Emotion-cause annotation (48 classes)	Comprehensive emotion-cause relationship mapping	Systematically removes emojis; no emoji-text emotional mapping
<i>EmoTag1200</i>	(Shoeb & de Melo, 2020)	1,200 emojis	Emoji-emotion intensity mapping	Covers Plutchik's 8 emotions with human ratings	Emoji-side associations only; no text congruence evaluation
<i>Sentiment of Emojis</i>	(Novak, Smailovic, Sluban, & Mozetic, 2015)	751 emojis	Emoji sentiment lexicon creation	First large-scale systematic emoji sentiment analysis	Only assesses emoji-side sentiment; no emotional pairing framework
<i>EmojiNet</i>	(Wijeratne, Balasuriya, Sheth, & Doran, 2017)	2,389 emojis	Semantic sense inventory	Comprehensive emoji-meaning mapping with API access	No emotion classification or congruence evaluation
<i>Emoji2Vec</i>	(Eisner, Rocktäschel, Augenstein, Bošnjak, & Riedel, 2016)	1,661 emojis	Unicode-based emoji embeddings	Avoids frequency bias through description-based learning	Based on formal descriptions rather than real-world usage patterns

Table 2.1 Comparative Analysis of Major Emotion-Emoji Systems

These limitations collectively highlight the critical need for a multimodal, context-aware evaluation framework, one that can assess not only whether an emoji fits a message textually or semantically, but whether it enhances or distorts the intended emotional tone. This gap in the literature forms the basis for the framework proposed in this study.

2.4 LLMs and Emotional Content Generation

The emergence of Large Language Models has significantly reshaped both the analysis and generation of emotional content. Yet, their ability to seamlessly integrate emojis with text remains an ongoing challenge. While modern transformer-based architectures excel at recognizing and producing a wide range of emotional patterns, achieving robust, nuanced multimodal integration, particularly for emoji-text relationships, remains a work in progress.

Recent studies illustrate the evolution of this field. The integration of BERT and RoBERTa with explicit emoji processing has led to demonstrable gains in emotional inference. By incorporating emojis into BERT-based architectures, a study (Khan, Majumdar, & Mondal, 2025) was able to achieve more refined emotion detection capabilities beyond what text alone provides. Similarly, (Talaat, 2023) observed that the hybrid DistilBERT-GLG, which employed targeted emoji preprocessing, yielded performance improvements in sentiment classification, suggesting the value of treating emojis as first-class emotional signals within model pipelines. These advancements emphasize that dedicated frameworks for emoji analysis can enrich transformer-based models, though the systematic assessment of their capabilities to understand or generate emotionally congruent emoji remains underexplored.

Multimodal attention mechanisms address emoji-text interaction challenges through sophisticated fusion approaches. (Yinxia, Zhou, Zhou, & Ji , 2024) developed mutual attention mechanisms combining BERT textual embeddings with ResNet-50 visual emoji features, achieving a 1.37% macro F1 and 2.30% accuracy improvements. (Tang , et al., 2024) demonstrated relative accuracy improvements of 2.3%, 10.9%, and 2.7% across three benchmark datasets when emojis were included using a cross-attention mechanism in their EMFSA model, suggesting that proper architectural design can capture emoji-text synergies effectively.

On the generative front, models like GPT-4 have shown promising advances in capturing the nuances of emoji use in real-world communication. Recent evaluations of GPT-4V indicate that it outperforms traditional approaches in contextual emoji interpretation and sentiment analysis, successfully navigating the subtleties of emoji use and cultural meaning in many cases. Nonetheless, even state-of-the-art LLMs continue to struggle in scenarios requiring fine-grained emotional differentiation or heightened cultural awareness, and systematic, context-sensitive assessment of their emoji-emotion alignment remains limited.

Recent advances in generative language models, particularly GPT-3.5, have extended the frontier of contextual emoji interpretation. (Zhou, et al., 2025) Conducted a systematic evaluation of ChatGPT's ability to decode emoji meanings across diverse social media contexts. Their findings indicated that GPT-3.5 outperformed GPT-4 on specific benchmarks, achieving 63.1% accuracy on an Arabic sarcasm detection task involving emoji use, compared to GPT-4's 60.1%. This suggests that even earlier-generation models, when appropriately prompted, can navigate emoji semantics with a level of consistency approaching that of human annotators. However, the study also highlights critical limitations. ChatGPT's interpretive accuracy deteriorates in culturally nuanced scenarios, especially when prompts lack explicit cultural framing. In such cases, the model often fails to generate accurate or contextually appropriate explanations. These results reinforce the idea that while LLMs possess broad lexical and symbolic knowledge of emojis, their effectiveness in real-world communication depends heavily on prompt specificity and cultural contextualization. This underscores the need for more robust frameworks to evaluate not just emoji interpretation, but their communicative impact when inserted back into digital discourse.

Chain-of-thought prompting, a recent advancement, enables models to elaborate on their reasoning process when assigning emotions or suggesting emojis. This approach holds promise for generating more contextually and emotionally resonant outputs by allowing the model to explain the emotional rationale behind emoji placement. However, the frameworks for evaluating the real-world communicative appropriateness of these reasoning-driven emoji suggestions are still in their early stages.

2.5 Current Approaches and Their Limitations

Existing methods for emoji-emotion analysis can be grouped into lexicon-based, machine learning, deep learning, and multimodal approaches. Lexicon-based systems, such as the Emoji Sentiment Ranking (Novak, Smailovic, Sluban, & Mozetic, 2015) and NRC Emotion Lexicon (Mohammad & Turney, 2013), primarily focus on capturing the sentiment aspect of communication—that is, broadly classifying emojis or words based on positive, negative, or neutral polarity. While computationally efficient and interpretable, these sentiment-oriented frameworks lack granularity in identifying specific emotional states or their intensities. Moreover, they treat emojis as having fixed emotional meanings, often ignoring how multi-faceted emoji interpretation is in real digital conversations.

Machine learning and deep learning models have advanced the field by learning emoji usage patterns directly from large datasets, thereby capturing more nuanced signals than static lexicons. DeepMoji (Felbo, Mislove, Søgaard, Rahwan, & Lehmann, 2017), for example, uses a massive corpus of tweets to learn emoji-aware representations, demonstrating improved performance on emotion-related tasks. However, such models generally optimize for accuracy in emoji prediction or sentiment classification without evaluating whether the predicted emojis are contextually appropriate or emotionally coherent when paired back with text.

Multimodal approaches integrate textual and emoji inputs, improving classification results by modeling their interactions. Despite these advances, evaluation frameworks predominantly rely on the overall accuracy or correlation metrics and rarely address whether emoji-text combinations maintain emotional coherence or appropriateness in communication. Additional challenges include cultural and demographic biases due to predominantly Western training data and the difficulty of adapting to the dynamic, evolving nature of emoji use across platforms.

Collectively, these limitations underscore the need for context-sensitive frameworks that extend beyond broad sentiment polarity and classification accuracy, to assess how emojis and text jointly convey layered, contextually appropriate emotional meaning.

2.6 Gaps in Current Research

The comprehensive analysis reveals a fundamental gap in existing research- the absence of context-aware evaluation frameworks that assess congruence between textual emotional content and emoji usage.

While significant progress exists in both text-based emotion analysis and emoji sentiment classification, their intersection remains critically underexplored.

Core gaps identified in current systems include:

- Absence of robust emotional coherence assessments: No prevailing model or dataset systematically evaluates how emojis and text together shape intended emotion.
- Multimodal blindness: Most models either ignore emojis entirely or fail to consider the emergent meanings produced by their interaction with text, even though, in many cases, emojis bear the main emotional load of brief digital messages.
- Persistent limitations exist in modeling the complexity of human affect. Most systems are restricted to recognizing only basic emotion categories, such as Plutchik's eight primary emotions, and are unable to capture compound, blended, or ambiguous emotional states found in natural communication.
- Over-reliance on context-insensitive evaluation metrics, at the expense of developing nuanced, human-centered assessment criteria for emotion-emoji congruence and communicative appropriateness.

By identifying these gaps, critically analyzing prior approaches, and positioning the need for a theory-driven, computationally robust evaluation framework, this chapter fulfills Objective O-1. It lays the conceptual foundation for the framework developed in Chapter 3, where Objective O-2 is addressed.

3 A Novel Framework for Evaluating Emoji Augmentation

Building on the research gaps identified in Chapter 2, this chapter addresses the central research question “Can LLMs effectively augment plain text with emojis?” by defining the methodological framework used to evaluate effectiveness. As outlined in Section 1.4.3, effectiveness is operationalized through four evaluation dimensions: Consistency (H1), Appropriateness (H2), Preservation (H3), and Interpretive Impact (H4).

Guided by these evaluation dimensions, this chapter introduces a dual-metric evaluation framework that integrates the Semantic Preservation Score (SPS) and the Emoji–Emotion Congruence (EEC) measure. Together, these metrics allow for independent yet complementary assessment of semantic fidelity and emotional alignment, avoiding the conflation of the two qualities and enabling deeper diagnostic insight. A dedicated Quality Assurance (QA) system ensures reliability and reproducibility across models, datasets, and prompting strategies.

The framework is grounded in Plutchik’s comprehensive emotion theory, enabling assessment of both basic and complex emotional categories, including secondary, tertiary, and blended dyads often overlooked in emoji-related research. It also embeds scalable computational pipelines that allow results to be compared across multiple LLM architectures and input types.

By defining the theoretical basis, computational architecture, metric design, and integration strategy for assessing emoji augmentation, this chapter fulfils Objective O-2. It translates the study’s conceptual aims into a concrete, testable structure, setting the stage for the experimental implementation in Chapter 4 and the hypothesis validation in Chapter 5.

3.1 Design Philosophy: Multi-Dimensional Effectiveness

Evaluating emoji augmentation requires rethinking traditional text-based quality metrics. Emojis introduce an inherently multi-dimensional layer to communication that involves both meaning and emotion. The section outlines the design principles and architecture that shape the proposed evaluation framework, ensuring both methodological rigor and applicability across varied communication contexts.

3.1.1 Foundational Design Principles

The framework is built on four guiding principles. Together, they address the analytical gaps identified in Chapter 2 and underpin the methodological contributions in this chapter.

Semantic–Emotional Duality Principle: Traditional evaluation metrics in natural language processing often assume that textual quality can be judged solely on semantic similarity (Papineni, Roukos, Ward, & Zhu, 2002; Lin, 2004). Emoji augmentation, however, demands more. It must preserve the original meaning while also expressing the intended emotional tone through well-chosen visual symbols. Achieving high-quality augmentation means meeting both requirements - semantic integrity and emotional alignment - so that the message is enhanced rather than distorted.

Multi-Method Convergence Principle: Emotional meaning is often interpreted differently by different readers (Russell, 1980; Barrett, 2006). Depending on a single evaluation metric can overemphasize one aspect of quality and miss others. To address this, evaluations should draw on several complementary analytical methods. When independent approaches point to the same conclusion, the results gain credibility while still reflecting the complexity of emotional communication.

Psychological Grounding Principle: Effective evaluation of emotional content should be grounded in established psychological theories of human emotion (Plutchik, 1980; Ekman, 1992). In this framework, the primary reference is Plutchik’s Wheel of Emotions, which maps the relationships between core emotions to capture the breadth of human emotional experience. Anchoring evaluation criteria in such validated models ensures that assessments reflect how emotions are actually structured, expressed, and interpreted, rather than relying solely on computational shortcuts.

Scalable Precision Principle: A practical framework must deliver consistent and interpretable results across different scales of analysis, from a single emoji-text pair to an entire corpus without losing precision (Fleiss, Levin, & Paik, 2003). It should balance the analytical depth needed for research with the efficiency required for real-time applications. This scalability ensures that the same evaluation criteria remain reliable, whether used for detailed investigation or large-scale deployment.

3.1.2 Architectural Framework Design

The evaluation framework operationalizes its foundational principles through a coordinated dual-metric architecture that jointly addresses the semantic and emotional dimensions of emoji augmentation quality.

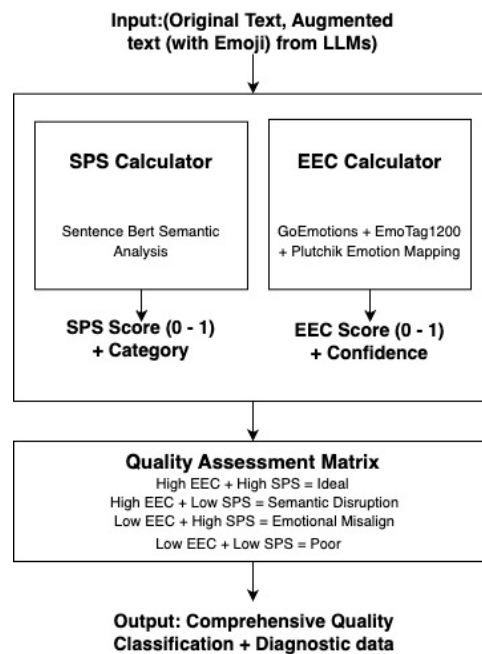


Figure 3.1 Simplistic SPS + EEC Framework

As illustrated in Figure 3.1, the architecture integrates two specialized scoring components, a multi-system integration module, and a quality assessment process that produces both categorical judgments and diagnostic insights.

Component	Primary Function	Input Data	Output Metrics	Integration Method
Semantic Preservation Score (SPS)	Quantify semantic integrity preservation	Original text, Augmented text	SPS score (0-1), Category	Sentence-BERT similarity analysis
Emoji Emotion Congruence Score (EEC)	Measure emotional alignment accuracy	Text emotions, Emoji emotions	EEC score (0-1), Confidence	Multi-system integration approach
Multi-System Integration	Coordinate existing emotion detection systems	Emoji symbols, Text content	Emotion vectors, Dyad relationships	GoEmotions + EmoTag1200 + PyPlutchik
Quality Assessment Module	Integrate scores into actionable metrics	SPS scores, EEC scores	Combined quality categories	Systematic evaluation synthesis

Table 3.1 Framework Component Overview

Semantic Preservation Score (SPS) Component

The SPS module addresses the core requirement that emoji augmentation must not compromise the semantic integrity of the original message. Although emojis are intended to enhance communicative effectiveness, inappropriate placement or selection can introduce ambiguity, shift emphasis, or alter meaning in ways that degrade message quality. The SPS quantifies the extent to which the original textual meaning is preserved after emoji augmentation by applying similarity analysis tailored to the unique properties of emoji-augmented text. This ensures that added expressiveness does not come at the expense of clarity or fidelity.

Emoji Emotion Congruence Score (EEC) Component

The EEC represents the principal methodological innovation of this research. It systematically evaluates the degree to which the emotions expressed by inserted emojis align with the emotional tone of the source text. The score is derived through a multi-method approach combining vector-based emotion representation, categorical emotion mapping, and intensity scaling, integrated from multiple established systems (GoEmotions, EmoTag1200, and PyPlutchik). This enables accurate assessment of both direct emotional matches and more nuanced, multi-emotion expressions.

Quality Category Classification System

By evaluating both SPS and EEC scores together, the framework can identify not only the overall quality of an emoji augmentation but also the specific nature of any issues.

This dual-metric approach addresses a key limitation of single-metric systems, which can only indicate whether quality is high or low but cannot reveal whether problems arise from changes in meaning, mismatched emotion, or both.

The resulting classification provides a direct link to improvement strategies: cases with low SPS but high EEC indicate a need to adjust the text to preserve meaning. In contrast, cases with high SPS but low EEC suggest changes to emoji selection. Table 3.2 defines the score thresholds, categories, interpretations, and recommended actions for each classification outcome.

EEC Score	SPS Score	Quality Category	Interpretation	Recommendation
High (≥ 0.7)	High (≥ 0.95)	Ideal	Excellent emotional alignment with semantic preservation	Accept augmentation
High (≥ 0.7)	Low (< 0.95)	Emotionally Accurate, Semantically Disruptive	Good emoji selection but text modification issues	Revise text preservation
Low (< 0.6)	High (≥ 0.95)	Semantically Preserved, Emotionally Misaligned	Good text preservation but inappropriate emoji selection	Revise emoji selection
Low (< 0.6)	Low (< 0.95)	Poor Quality	Neither emotional accuracy nor semantic preservation achieved	Reject augmentation

Table 3.2 Quality Category Classification Matrix

This integrated architecture supports both automated augmentation pipelines and human evaluation processes. It provides interpretable, criteria-based quality judgments and clear improvement guidelines, making it applicable to both research and practical deployment contexts. By maintaining dual attention to semantic preservation and emotional congruence, the framework delivers a more comprehensive and reliable evaluation of emoji augmentation quality than single-dimensional approaches.

3.2 Semantic Preservation Score Development

The Semantic Preservation Score (SPS) addresses a basic but essential requirement that emoji augmentation should add expressiveness without altering the intended meaning of the original message. This section explains the theoretical background, the reasoning behind the method, and the approach used to calculate SPS, which forms the first part of the dual-metric evaluation framework.

3.2.1 Theoretical Foundation and Literature Integration

SPS is grounded in distributed semantic theory, which models linguistic meaning as points or vectors in a high-dimensional space. In this representation, semantic similarity corresponds to the proximity of these vectors: the closer the vectors, the greater the semantic overlap (Mikolov, Sutskever, Corrado, & Chen, 2013; Devlin, Chang, Lee, & Toutanova, 2019). This approach is widely used in natural language processing and provides a solid basis for measuring how much of the original meaning is preserved after modification.

Applying these principles to emoji augmentation requires addressing the distinctive properties of emoji-enhanced communication. Unlike paraphrasing or translation, where the task involves replacing one textual form with another, emoji augmentation adds visual symbols alongside the existing text. Ideally, these additions enrich emotional tone while leaving the underlying semantic content intact. The SPS framework is therefore designed to detect whether such augmentation maintains the intended meaning or inadvertently introduces semantic drift.

3.2.2 Methodological Architecture

SPS is calculated using several complementary checks that together give a complete picture of meaning preservation:

- *Semantic similarity* - Measures how close the meanings of the original and augmented texts are using sentence embeddings (e.g., Sentence-BERT).
- *Lexical and character overlap* - Compares words and characters to find any unintended textual changes.
- *Structural checks* – Looks at changes in length, word count, or sentence structure that may signal a shift in meaning.
- *Perfect preservation flag* – Notes when the augmented text (without emojis) exactly matches the original, which is the ideal case.

These elements work together to detect both small and significant shifts in meaning. Algorithm 3.1, presented later, shows the whole calculation process and how each part contributes to the final SPS value. This approach makes SPS both sensitive enough to

catch subtle changes and stable enough for large-scale evaluation, allowing it to be used confidently in both automated systems and human-led assessments of emoji-augmented communication.

ALGORITHM: CalculateSPS(original_text, augmented_text)

INPUT:

- *original_text (string),*
- *augmented_text (string)*

OUTPUT:

- *SPS score (float 0-1)*
- *category (string)*
- *diagnostics (dict)*

BEGIN

1. *Remove emojis from augmented_text; normalise whitespace in both texts.*
2. *If cleaned texts match exactly, return SPS = 1.0 and category = "perfect".*
3. *Encode both texts with Sentence-BERT; compute cosine similarity for semantic match.*
4. *Compute character preservation using normalised Levenshtein distance.*
5. *Calculate penalties for word count change, length change, and character change.*
6. *Multiply semantic similarity, character preservation, and penalty adjustments; limit to [0, 1].*
7. *Assign category based on final score.*
8. *Return score, category, and component metrics as diagnostics.*

END

ALGORITHM 3.1 SPS Calculation

The semantic similarity component uses Sentence-BERT embeddings (Reimers & Gurevych, 2019) to compute the cosine similarity between the original text and the emoji-cleaned version of the augmented text. This method captures nuanced semantic relationships while remaining resilient to superficial wording differences that do not alter the core meaning.

Sentence-BERT is selected over traditional word embedding models for several reasons:

- It generates sentence-level representations rather than isolated word vectors, enabling it to model meaning in context.

- It is fine-tuned on natural language inference and semantic textual similarity tasks, which improves its ability to detect subtle shifts in meaning.
- It also supports robust comparison of variable-length inputs, making it suitable for emoji augmentation scenarios where message length and structure can vary.

These qualities make Sentence-BERT well-suited for accurately assessing semantic preservation in cases where even minor contextual changes could affect interpretation.

The calculations begin by preprocessing the inputs to ensure a fair like-for-like comparison. Emojis are removed from the augmented text so that only the written content remains, and whitespace is normalized in both texts. A quick, perfect preservation check compares the cleaned, augmented text to the original. If they are identical, the algorithm returns an SPS of 1.0 and labels the case *Perfect*, since the augmentation has added only emojis and made no textual changes. When differences exist, the algorithm measures the semantic similarity at the sentence level using Sentence-BERT embeddings (Reimers & Gurevych, 2019).

Let $E_{original}$ and $E_{augmented}$ be the sentence embeddings for the original text and the emoji-cleaned augmented text, respectively. Their cosine similarity provides the semantic match, SS:

$$SS = \frac{E_{original} \cdot E_{augmented}}{\|E_{original}\| \cdot \|E_{augmented}\|}$$

As high semantic similarity can sometimes mask heavy rewriting, the algorithm also computes a character-level preservation measure using normalized edit distance. With $Levenshtein(\cdot)$ denoting the edit distance, i.e., the minimum number of single-character insertions, deletions, or substitutions needed to transform the original text into the augmented version, and all lengths measured in characters, the Character Preservation (CP) score is calculated as:

$$CP = 1 - \frac{Levenshtein(original, augmented)}{\max(len(augmented), len(original))}$$

This component detects structural changes that could affect tone, nuance, or clarity, even when the overall meaning remains intact.

To further safeguard against over-editing, the framework applies modification penalties. These target three areas:

- (i) The word count penalty WCP, which triggers if token differences exceed two words;

$$WCP = \min(0.2, \text{no. of words differences}),$$

- (ii) The length change penalty LCP, which applies when text length changes by more than 20%.

$$LCP = \min(0.15, \text{length_ratio} - 1),$$

- (iii) The character change penalty, CCP, which is proportional to the reduction in character preservation.

$$CCP = \text{MIN}(0.25, (1 - \text{char_preservation}) \times 0.4)$$

Based on preliminary testing, small variations in word count or text length were found, which often resulted from benign adjustments, such as correcting spacing or rephrasing for fluency, grammatical, or spelling checks, rather than from changes that altered meaning. To accommodate this, tolerance levels were set at ± 2 tokens for word count and $\pm 20\%$ for text length, allowing such minor edits without penalty. These thresholds are in line with established evaluation practices, such as BLEU's brevity penalty, which accommodates modest length differences while safeguarding semantic fidelity (Papineni, Roukos, Ward, & Zhu, 2002).

The use of bounded penalties: 0.20 for word count, 0.15 for length, and 0.25 for character preservation ensures that no single factor disproportionately lowers the SPS score. This approach follows best practices from edit-distance-based evaluation frameworks, which recommend proportional rather than absolute penalization to maintain balanced measurement across multiple dimensions (Levenshtein, 1965; Popović & Ney, 2011).

The final SPS is obtained by integrating all components multiplicatively, ensuring that a significant drop in any one dimension reduces the overall score. This composite score is constrained to the range $[0,1]$:

$$SPS = \min(1.0, \max(0.0, SS \times CP(1 - WCP) \times (1 - LCP) \times (1 - CCP)))$$

This multiplicative design ensures that high semantic similarity cannot fully compensate for structural degradation, while still being sensitive to subtle preservation issues.

Once calculated, the SPS is mapped to qualitative categories that facilitate interpretation and guide decision-making. Table 3.3 defines the categorization thresholds and their corresponding implications, ranging from *Perfect* preservation to *Poor* quality. Alongside the score and category, the algorithm outputs a diagnostic breakdown of component metrics, enabling evaluators to identify whether a reduction in score is driven by semantic drift, excessive rewriting, or structural imbalance.

Score Range	Category	Interpretation	Communication Impact	Recommended Action
1.0	Perfect	Identical text with emoji addition only	No semantic impact	Accept
0.95-0.99	Excellent	Minimal modification with high preservation	Negligible impact	Accept
0.90-0.94	High	Good preservation with minor changes	Minor clarity reduction	Accept with monitoring
0.85-0.89	Moderate	Acceptable preservation with noticeable changes	Moderate clarity impact	Review modification necessity
0.80-0.84	Low	Significant drift but core meaning preserved	Substantial clarity reduction	Consider revision
<0.80	Poor	Major alteration or semantic corruption	Severe communication degradation	Reject

Table 3.3 SPS Score Categories and Interpretations

3.3 Emoji Emotion Congruence Score Framework

The Emoji Emotion Congruence (EEC) Score represents the principal methodological innovation of this research. It provides a systematic approach to quantifying the degree of emotional alignment between the textual message and its emoji augmentation. Contrasting to single-label matching, the EEC framework evaluates both direct emotional correspondence and more complex affective relationships, making it sensitive to nuanced, context-dependent meanings in human–emoji communication.

3.3.1 Comprehensive Pipeline Architecture

The EEC framework operates through a multi-stage pipeline (Figure 3.2) that processes original text and emojis extracted from augmented text in separate, parallel channels. This separation ensures that textual emotion detection is based solely on the original content, while emoji emotion detection focuses exclusively on the symbols added during augmentation.

Processing is conducted in parallel:

- *Text path* → Original unmodified text is processed to derive its emotional profile.
- *Emoji path* → The augmented text is stripped of its emojis, and the extracted emojis are independently analyzed for their emotional properties.

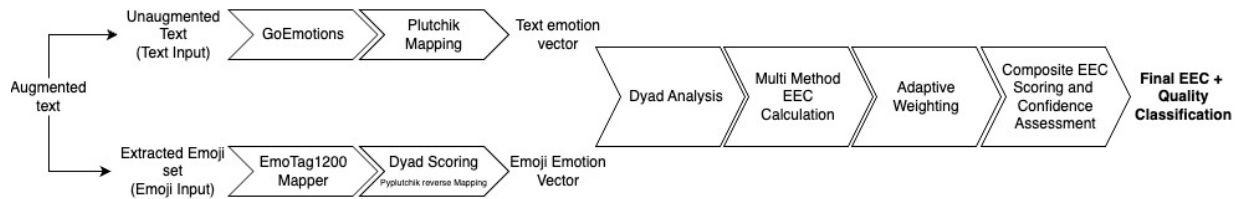


Figure 3.2 EEC Pipeline Architecture

Once both profiles are obtained, the system compares them through a multi-method calculation that integrates vector similarity, categorical relationships, and intensity matching

3.3.2 Emotion Extraction and Vector Mapping

The text pathway applies the GoEmotions classifier (Demszky, et al., 2020), which produces probabilistic emotion scores. These scores are then mapped to Plutchik’s eight primary emotions: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation through a pre-defined mapping table, yielding a normalized eight-dimensional emotion vector.

The emoji pathway uses EmoTag1200 (Shoeb & de Melo, 2020) to detect probable emotional categories associated with each emoji. The resulting outputs are also mapped

into Plutchik’s eight-emotion space, ensuring that both text and emoji representations exist in the same conceptual framework for direct comparison.

By using identical mapping schemes for both channels, the framework ensures that the resulting emotion vectors are directly comparable, forming the foundation for subsequent dyad analysis and metric computation.

3.3.3 Dyad Generation and PyPlutchik

After both text and emoji emotion vectors are mapped into Plutchik’s eight-dimensional space, they are further processed to identify emotional dyads combinations of two emotions that occur together in Plutchik’s model.

Dyad analysis in the EEC framework captures the structured relationships between Plutchik’s primary emotions, representing blended states such as *Joy + Trust* → *Love* or *Fear + Surprise* → *Awe*. These are grouped into four classes: *Primary*, *Secondary*, *Tertiary*, and *Opposite*, each assigned a weighting that reflects its psychological significance.

The system uses PyPlutchik (Semeraro, Vilella, & Ruffo, 2021) as a reference model to perform a reverse mapping from the eight-dimensional emotion vectors (from both text and emoji channels) into the whole dyadic space. For each predefined dyad, an activation score is assigned as the minimum intensity of its two constituent emotions, provided both exceed a small activation threshold (e.g., 0.05). Active dyads are then weighted according to their category.

These dyad representations serve three purposes within the EEC pipeline:

1. Enhancing categorical matching by rewarding overlapping dyads and penalizing strongly contradictory pairs.
2. Quantifying emotional complexity by summing weighted dyad counts (see Section 3.3.4).
3. Guiding adaptive weighting, where richer dyad structures increase the influence of categorical matching, while simpler, high-polarity profiles favor cosine similarity.

This reverse-mapping approach enables the framework to model the complete emotional structure of both text and emojis, identifying subtle alignments or divergences that would remain undetected using only single-label or vector-based comparisons.

3.3.4 Calculation of Core Emotional Metrics

From each Plutchik emotion vector, four core metrics are calculated for both text and emoji. Given an Emotional vector E associated with a text or an Emoji,

Polarity – Measures the degree to which a single primary emotion dominates an emotional profile. High polarity indicates a clear, focused emotional signal, while low polarity suggests a diffuse or mixed state. It is calculated as:

$$Polarity = \frac{\max(E)}{\sum E}$$

Balance – Measures how evenly emotions are distributed across all eight dimensions. Perfect balance occurs when all emotions are equally represented; low balance indicates concentration in a small subset of emotions.

$$Balance = 1 - \frac{Variance(E)}{Mean(E)^2}$$

Intensity – Measures the total magnitude of emotional expression. Helps assess the strength of the emotional expression, ensuring the emoji's emotional force matches that of texts.

$$Intensity = \sum E$$

Complexity – Measures the richness of blended emotional states, calculated as the weighted sum of active dyads. It helps assess the richness and diversity of blended emotional states, capturing nuanced combinations beyond single primary emotions.

$$Complexity = D_p + D_s * 0.8 + D_t * 0.6 + D_o * 1.2$$

where D_p , D_s , D_t and D_o are the counts of active Primary, Secondary, Tertiary, and Opposite dyads, respectively. These metrics are consistently applied to both channels, ensuring that emotional properties are assessed on the same basis before being integrated into the final EEC computation.

Dyad Category	Count	Emotional Distance	Example Dyads	Psychological Significance
Primary	8	Adjacent on wheel	Joy + Trust → Love, Fear + Surprise → Awe	Basic emotional combinations
Secondary	8	One position apart	Joy + Fear → Guilt, Trust + Surprise → Curiosity	Moderate complexity relationships
Tertiary	8	Two positions apart	Joy + Sadness → Bittersweetness	High complexity relationships
Opposite	4	Directly opposing	Joy ↔ Sadness, Trust ↔ Disgust	Maximum complexity contradictions

Table 3.4 Complete Dyad Classification System

The weighting scheme ensures that common but meaningful blends (Primary) are not undervalued, while rare but cognitively demanding constructs (Opposite) receive proportionally greater influence on the score. Conversely, blends with higher emotional distance but lower communicative salience (Tertiary) are weighted less heavily, avoiding overestimation of complexity in cases where emotional variety is high but not necessarily meaningful in everyday interpretation.

3.3.5 Multi-Method EEC Calculation System

The EEC framework evaluates emotional congruence using three complementary analytical methods. Each method captures a different facet of text–emoji emotional alignment, ensuring both precision and interpretability in the final evaluation.

Method 1: Enhanced Cosine Similarity

This method measures the geometric alignment between the text and emoji emotional profiles, each represented as an eight-dimensional vector in Plutchik’s emotion space. Cosine similarity is used to calculate the base score, reflecting the angular closeness of these vectors. This ensures that high similarity values are achieved only when both profiles share a similar pattern of emotional composition. The base similarity score is refined with targeted adjustments to reflect higher-order emotional properties.

- Polarity bonus: rewards clear, focused emotional expression in emojis.
- Balance bonus: rewards even emotional distribution.
- Dyad complexity bonus: rewards the richness of blended emotions.

A magnitude penalty is applied when total intensities differ significantly, preventing inflated similarity scores in cases where the strength of expression is mismatched. This method is well-suited to capturing general emotional alignment even when exact dominant emotions differ. Dyadic complexity values used in this method are derived solely from the emoji emotion vector.

INPUT:

- *Text*
- *Emoji*
- *emoji_polarity,*
- *emoji_balance,*
- *dyad_counts*

OUTPUT:

- *cosine_result*

BEGIN

1. *Convert both emotion profiles to eight-dimensional vectors.*
2. *If either vector has zero magnitude, return similarity = 0.*
3. *Compute base cosine similarity between vectors.*
4. *Apply polarity bonus if emoji_polarity > threshold.*
5. *Apply balance bonus if emoji_balance > threshold.*
6. *Compute dyadic complexity bonus based on weighted dyad counts.*
7. *Apply magnitude penalty if intensity ratio exceeds tolerance limits.*
8. *Calculate adjusted similarity:*

$$\text{adjusted} = \text{base} + \text{bonuses} - \text{penalties}$$
9. *Clamp adjusted similarity between 0 and 1.*
10. *RETURN cosine_base_similarity, adjusted_similarity, polarity_bonus, balance_bonus, dyad_complexity_bonus, magnitude_penalty.*

END

ALGORITHM 3.2 EEC - Enhanced Cosine Similarity Calculation

This approach reflects the conceptual design of the EEC framework, in which the text provides the emotional baseline and the emoji channel serves as the augmentation under evaluation. By restricting dyad complexity bonuses to emojis, the method rewards only those complex emotional combinations that are introduced or reinforced by emoji use, rather than those already inherent in the text.

Method 2: Advanced Categorical Matching

This method performs a structured, multi-layered comparison of emotional content, extending far beyond simple label matching. Unlike cosine similarity, which operates on the overall vector shape, this method captures interpretable relationships at the categorical level. It evaluates the relationship between the text's primary emotion and that of the emojis through family analysis (shared positive or negative valence), arousal assessment (high- or low-energy emotional states), and secondary emotion alignment.

Exact matches between dominant emotions receive the highest score, while emotions in the same positive or negative family, or those sharing similar arousal levels, earn reduced credit. Secondary matches are scored lower to reflect weaker alignment.

Component	Score Range	Criteria	Enhancement Factor
Exact Match	1.0	Identical dominant emotions	None (perfect score)
Family Match	0.7	Same emotional family (+ve/-ve)	Arousal consistency
Arousal Match	0.5	Compatible arousal levels (high/low)	Family alignment
Secondary Match	0.3	Secondary emotion alignment	Cross-matching bonus
Dyad Enhancement	+0.15	Text emotion in emoji dyads	Opposite dyad bonus

Table 3.5 Categorical Matching Score Components

The dyad enhancement component checks whether the text's primary emotion appears in any emoji dyad, rewarding the matches that convey richer or more complex emotional relationships, including opposing emotions (e.g., joy vs sadness).

Dyads are applied only to emojis, as their role is to add nuance or contrast to the text's existing tone rather than mirror it. This distinction isolates the augmentative effect of emojis, enabling the framework to recognize when they enrich, diversify, or refine the expressed emotion without conflating these effects with textual complexity.

By combining structured categorical logic with selective dyad integration, this method captures relationships that vector similarity alone might miss, while ensuring transparency for analysis and evaluation.

Method 3: Enhanced Intensity Analysis

The Enhanced Intensity Analysis evaluates how closely the magnitude of emotional expression in the emoji channel aligns with that of the text channel. This method incorporates emotion-specific psychological weights, ensuring that comparisons reflect the communicative and evolutionary salience of each emotion. For instance, fear receives a higher weight than trust due to its greater significance in human affective responses.

Emotion	Weight	Psychological Rationale	Impact on Analysis
Joy	1.0	Baseline positive emotion	Standard processing
Anger	1.1	High arousal, strong impact	Enhanced weight
Fear	1.2	Evolutionary significance	Maximum weight
Sadness	0.9	Lower arousal, contemplative	Reduced weight
Trust	0.8	Social bonding emotion	Reduced weight
Disgust	1.0	Avoidance emotion	Standard processing
Surprise	0.9	Brief, transient emotion	Reduced weight
Anticipation	0.8	Future-oriented emotion	Reduced weight

Table 3.6 Emotion-Specific Weights for Intensity Analysis

The framework compares these weighted intensities between text and emoji, classifying their relationship into one of four categories:

- *Well-matched*: Intensities differ by less than 20%, indicating proportional emotional alignment.
- *Emoji-strong*: Emojis express significantly stronger emotions, suggesting amplification or overexpression.
- *Text-strong*: Text expresses stronger emotion, indicating potential underrepresentation by emojis.
- *Mismatched*: Large discrepancies signal poor emotional alignment.

The weights reflect the established findings in affective science. High-arousal emotions such as fear and anger exert a much stronger perceptual and behavioral influence than lower-arousal states like trust or anticipation (Russell, 1980; Kensinger, 2009; Ohman, 2009). Prioritizing high-impact emotions prevents overestimating congruence when key salient emotional cues are mismatched, aligning the analysis with both psychological theory and affective computing practice (Calvo & D'Mello, 2010).

By weighting emotions and applying empirically defined thresholds, this analysis avoids overestimating congruence in cases where one modality disproportionately amplifies or diminishes expression. The result is a more accurate, psychologically grounded evaluation of how well emoji augmentation maintains proportional emotional intensity.

3.3.6 Adaptive Weighting and Composite EEC Scoring

The EEC framework combines cosine similarity, categorical matching, and intensity analysis into a single composite score. To ensure this score reflects the most contextually appropriate assessment, an adaptive weighting mechanism adjusts the contribution of each method based on the emotional characteristics of the input.

Three key factors guide Weighting adjustments:

1. *Emoji Polarity*: When polarity exceeds 0.7, emotional clarity is high, and greater weight is given to cosine similarity, as vector-based matching becomes more dependable.
2. *Emoji Balance*: A balanced emotional profile (balance > 0.8) produces a stable emotional signal, leading to a more even distribution of weights across all methods.
3. *Dyad Complexity*: A rich presence of dyads increases the weight assigned to categorical matching, prioritizing relational and structural emotion analysis over magnitude alone.

This adaptive approach is grounded in principles of multi-criteria decision-making (MCDM), which modify the relative importance of evaluation criteria based on situational relevance to improve decision quality (Triantaphyllou, 2000). Computationally, it parallels ensemble learning strategies (Dietterich, 2000), where multiple specialized models contribute to a final decision, with influence weighted by their context-specific reliability. In affective computing, this is consistent with established best practices that adjust feature importance according to clarity, complexity, and modality-specific constraints (Picard, 1997). By dynamically adapting weight allocation, the system ensures that the final composite score prioritizes the most reliable analytical perspective for each evaluation case while preserving transparency and interpretability.

INPUT:

- emoji_polarity (float), emoji_balance (float), dyad_counts (dict)

OUTPUT:

- adjusted_weights (dict),
- adjustment_reasons (list)

BEGIN

1. Initialise base weights:
 weights $\leftarrow$ {cosine: 0.4, categorical: 0.35, intensity: 0.25}
 adjustment_reasons $\leftarrow$ []
2. Polarity adjustment:
 IF emoji_polarity > 0.7 THEN
 weights.cosine += 0.1
 weights.categorical -= 0.05
 weights.intensity -= 0.05
 adjustment_reasons.ADD("high_polarity")
3. Balance adjustment:
 IF emoji_balance > 0.8 THEN
 weights $\leftarrow$ {cosine: 0.4, categorical: 0.3, intensity: 0.3}
 adjustment_reasons.ADD("high_balance")
4. Dyad richness adjustment:
 total_dyads $\leftarrow$ SUM(dyad_counts.values())
 IF total_dyads > 2 THEN
 weights.categorical += 0.05
 weights.cosine -= 0.03
 weights.intensity -= 0.02
 adjustment_reasons.ADD("rich_dyads")

 IF dyad_counts.opposite > 0 THEN
 weights.categorical += 0.03
 weights.intensity -= 0.03
 adjustment_reasons.ADD("complex_emotions")
5. Normalise weights:
 weight_sum $\leftarrow$ SUM(weights.values())
 FOR each key IN weights:
 weights[key] $\leftarrow$ weights[key] / weight_sum
6. RETURN weights, adjustment_reasons

END

ALGORITHM 3.3 EEC – Adaptive Weight Adjustment

The composite EEC score is calculated as:

$$EEC_{composite} = (w_{cosine} * S_{cosine}) + (w_{categorical} * S_{categorical}) + (w_{intensity} * S_{intensity})$$

where w_{cosine} , $w_{categorical}$, and $w_{intensity}$ are adaptively determined weights that sum to 1.0.

The final composite EEC score represents a context-sensitive, adaptively weighted measure of emotional congruence between text and emoji, combining geometric, categorical, and intensity-based perspectives into a single interpretable value.

The variance between method scores determines confidence in the result: lower variance indicates higher reliability. This confidence score, combined with the EEC composite score, is then mapped to a quality classification scale, facilitating interpretable outcomes for both research and applied evaluation contexts.

3.3.7 Confidence Assessment and Quality Classification

The final stage of the EEC framework addresses two critical requirements:

1. Quantifying the reliability of the composite EEC score through a confidence metric.
2. Translating the numerical score into interpretable quality categories for practical use.

Confidence Assessment

The confidence metric evaluates the agreement between the three constituent methods: cosine similarity, categorical matching, and intensity analysis. The underlying assumption is that when these methods produce highly similar scores, the final composite score is more trustworthy. Conversely, a wide variation between methods indicates uncertainty and reduces confidence.

Confidence is computed as:

$$Base\ Confidence = 1.0 - \min(0.4, \sigma(S_{cosine}, S_{categorical}, S_{intensity}))$$

where $\sigma(.)$ represents the standard deviation of the method scores.

A low deviation indicates high agreement and thus higher reliability. This is further refined by incorporating dyad richness, with additional weight given when multiple dyads are present:

$$\textbf{Final Confidence} = \textbf{min}(1.0, (\textbf{Base Confidence} + \textbf{min}(0.1, \textbf{Total Dyads} * 0.02)))$$

This approach ensures that confidence reflects both statistical consistency and qualitative depth in emotional analysis.

Quality Classification

The composite EEC score, alongside dyad complexity and opposite-dyad activation, determines the final quality classification (Table 3.7). The system uses a tiered structure to categorize outputs into “Excellent,” “High,” “Moderate,” “Low,” or “Very Low,” with modifiers applied to capture additional emotional sophistication.

Modifiers indicate cases of special interest where intense dyadic richness or complex, contradictory emotional configurations are present, even when the base score alone might suggest a lower classification.

Score Range	Base Classification	Rich Dyad Modifier	Complex Emotion Modifier	Validation Alignment
≥ 0.85	Excellent	+ rich_dyads (≥ 3 dyads, score ≥ 0.6)	+ complex_emotions (opposite dyads, score ≥ 0.5)	Exceeds appropriateness threshold
0.70 – 0.84	High	+ rich_dyads (≥ 3 dyads, score ≥ 0.6)	+ complex_emotions (opposite dyads, score ≥ 0.5)	Above appropriateness threshold
0.60 – 0.69	Moderate	+ rich_dyads (≥ 3 dyads, score ≥ 0.6)	+ complex_emotions (opposite dyads, score ≥ 0.5)	Meets threshold
0.40 – 0.59	Low	—	—	Below appropriateness threshold
< 0.40	Very Low	—	—	Significantly below threshold

Table 3.7 EEC Quality Classification System

This classification scheme builds on the adaptive weighting principles described in Section 3.3.6 and the dyad-based analysis framework introduced in Section 3.3.4. Incorporating dyad complexity as a modifier ensures that “rich” emotional compositions

are only recognized when both the number of dyads and their weighted significance exceed a meaningful threshold, preventing incidental matches from inflating quality scores. Likewise, the complex emotion modifier recognizes deliberate affective sophistication, such as opposing emotional states, only when their activation level is sufficiently high to indicate intentional communicative use.

From an academic standpoint, this approach is also like MCDM frameworks (Triantaphyllou, 2000), where both quantitative metrics and contextual indicators inform the final classification. It also follows established practices in affective computing (Picard, 1997), where reliability is improved by combining multiple complementary measures and weighting them by contextual relevance. The inclusion of confidence as a core output ensures that high-quality classifications are both numerically robust and methodologically defensible, supporting transparent evaluation in both research and applied settings.

In practice, this classification system allows the EEC framework to serve both diagnostic and evaluative functions. For example, an “Excellent” classification with a high confidence score signals not only strong emoji–text alignment but also high analytical certainty, making it suitable for use in automated quality control. Conversely, a “Moderate” score with low confidence may prompt targeted review, especially if it lacks dyadic richness or contains intensity mismatches. This dual focus on what the score is and *how reliable* it is ensures that the system is both actionable and interpretable.

This dual approach of quantifying confidence and categorizing quality ensures that the EEC framework produces results that are both statistically grounded and operationally interpretable, balancing computational precision with human usability.

3.4 Quality Assurance Framework

The implementation of the evaluation framework incorporates a dedicated Quality Assurance architecture designed to ensure both analytical consistency and methodological validity.

These measures guarantee that the system’s assessments of semantic preservation and emotional congruence remain stable, interpretable, and aligned with established research in affective computing and computational linguistics.

3.4.1 Quality Assurance Methodology

The QA process monitors the framework’s performance across multiple evaluation dimensions, combining semantic similarity measures, emotional alignment thresholds, and statistical stability indicators.

QA Dimension	Monitoring Metric	Target Value	Assessment Method	Academic Basis
EEC Consistency	Coefficient of variation	$CV < 0.25$	Variance analysis within emotion groups	Stability metrics in affective computing (Cambria, Poria, Gelbukh, & Thelwall, 2017)
EEC Appropriateness	Mean EEC score	≥ 0.6	Three-method ensemble evaluation	Agreement thresholds in multimodal sentiment analysis (Plutchik, 1980)
SPS Preservation	Mean SPS	≥ 0.95	Sentence-BERT similarity	High-similarity benchmarks in text transformation (Iyyer, Wieting, Gimpel, & Zettlemoyer, 2018)
Performance Correlation	EEC–emotion classification correlation	$r > 0.7$	Downstream task correlation analysis	Feature-performance correlation studies (Triantaphyllou, 2000; Dietterich, 2000)

Table 3.8 Comprehensive Quality Assurance Framework

The Semantic Preservation Score (SPS) threshold is grounded in established NLP best practices, where a cosine similarity score above 0.95 reliably indicates minimal semantic drift (Reimers & Gurevych, 2019).

Similarly, the $EEC \geq 0.6$ criterion reflects thresholds used in affective computing to represent “acceptable emotional alignment” between modalities (Cambria, Poria, Gelbukh, & Thelwall, 2017; Plutchik, 1980).

The EEC–Performance correlation is calculated by pairing each text sample’s *Emotion-Emoji Congruence (EEC)* score with its corresponding classification accuracy from a fine-grained automated emotion detector. Pearson’s correlation coefficient is then computed across emotion categories, with values of $r \geq 0.7$ indicating that stronger emoji–text

alignment tends to improve automated emotion recognition. This metric validates H4, showing that EEC is not only a measure of internal congruence but also a predictor of functional gains in AI-driven emotion interpretation. The finding is consistent with prior work in affective computing and NLP, where affective cues such as emojis have been shown to enhance recognition accuracy.

3.4.2 Analytical Infrastructure and Capabilities

The framework's architecture supports detailed diagnostics, statistical testing, and large-scale research analysis:

- Diagnostic Output Generation: Each evaluation produces method-specific scores (cosine similarity, categorical matching, intensity analysis), dyad complexity metrics, adaptive weighting coefficients, and quality classifications. This mirrors explainable AI principles by ensuring all results can be traced and validated.
- Statistical Analysis Support: The structured output facilitates variance analysis, correlation testing, and longitudinal performance monitoring, enabling both single-case evaluations and dataset-wide studies.
- Research Analysis Capabilities: The system supports targeted investigations such as emotion-specific performance trends, the impact of polarity and complexity on scoring, and inter-method consistency checks.

By combining empirically validated similarity measures, psychological emotion models (Plutchik, 1980), and multi-criteria decision-making principles, the QA framework ensures that results are not only computationally consistent but also psychologically interpretable. This dual validation of computational stability and theoretical grounding provides a strong basis for both research reliability and real-world deployment.

4 Experimental Design and Implementation

This chapter outlines the methodology developed to empirically evaluate whether large language models (LLMs) can effectively augment emotionally rich text with emojis, enhancing emotional interpretation while preserving semantic integrity. The study addresses challenges in dataset construction, experimental configuration, and evaluation strategy to generate robust evidence about the factors influencing emoji augmentation quality.

4.1 Research Design Overview

Recent advancements in large language models have demonstrated their capacity to learn from minimal examples across a wide range of natural language processing tasks (Brown, et al., 2020), as well as their utility as general-purpose text generation systems (Qin, et al., 2023). These capabilities provide a foundation for investigating whether LLMs can be effectively used to augment emotion-rich text with emojis, enhancing interpretive clarity while preserving the original semantic content.

To explore this question, the study adopts a structured, multi-variable experimental design that combines three LLMs with four distinct prompting strategies. This design enables systematic assessment of performance variation across models and the impact of prompting style on the emotional and semantic quality of emoji enhancement. Employing a factorial design, model types and prompt styles are varied across a standardized dataset to evaluate their individual and interactive effects on emoji enhancement outcomes. This approach ensures a thorough analysis of how specific configurations influence both the consistency and interpretability of the generated outputs. The section also details the rationale behind model and prompt selection, showing how each aligns with the study's central aim: to evaluate how well LLM-generated emoji augmentation.

The experimental configuration uses a factorial design matrix, resulting in a total of 24,000 observations. This thorough design supports rigorous statistical analysis and also helps ensure fair comparisons across different model types and prompting methods, accounting for potential confounding variables (Liang, et al., 2023).

4.2 Dataset Selection and Preparation

4.2.1 Dataset Selection Methodology

The dataset preparation process follows a rigorous four-stage course designed to create a balanced, representative corpus suitable for emoji augmentation. It ensures the quality and validity of the data and experiment. Figure 4.1 illustrates this multistage process. It begins with a broad set of candidate datasets. It then applies successive refinement steps to arrive at a final selection that meets the study's quality and representativeness, thereby fulfilling Objective O-3 of this research.

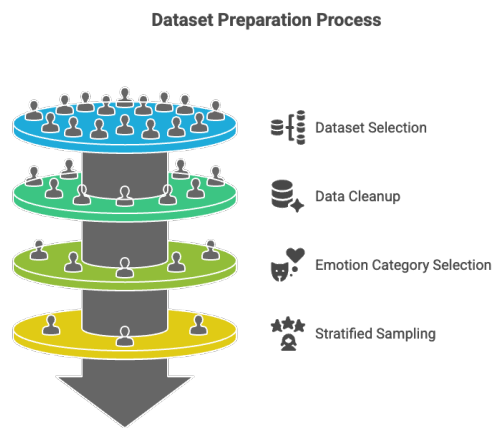


Figure 4.1 Dataset Preparation Process

The initial stage involved a systematic evaluation of the available emotion datasets. This evaluation was based on several criteria, such as diversity of emotional categories, depth of semantic detail, and whether the data structure would support meaningful emoji augmentation. Table 4.1 summarizes the main datasets reviewed, alongside their characteristics and the rationale for their selection or exclusion.

Dataset	Records	Emotions	Emoji Status	Platform	Year	Selection Decision
GoEmotions	58,000	27+1	Mixed	Reddit	2020	Not selected - platform bias
DeepMoji	1.2M	3	Present	Twitter	2017	Not selected - limited emotions
EMO-KNOW	700,000	48 (only 27 was visible in their dataset)	Emoji-free (<0.5% had some emoji)	Twitter	2022	Selected
SemEval-2018	22,000	4	Mixed	Multiple	2018	Not selected - limited scale

Table 4.1 Dataset Comparison Analysis

Upon careful comparison, the EMO-KNOW dataset was selected as the best fit. Its strengths were several:

- i. Provides an emoji-free baseline text, which is ideal for controlled augmentation experiments.
- ii. Captures real-world emotional expression through self-reported tweets.
- iii. Offers a broad and nuanced range of emotion categories.
- iv. Possess high-quality, large-scale annotations
- v. Data is both recent and relevant, having been compiled through 2022.

These features made EMO-KNOW the most robust and appropriate choice to support the experimental objectives and to ensure the validity of subsequent findings.

4.2.2 Data Cleaning and Preprocessing Pipeline

To warrant data quality and consistency throughout the experiment, the dataset underwent another structured four-stage cleaning and preprocessing pipeline. As shown in Figure 4.2, this stepwise process helped remove inconsistencies, standardize the formats, and preserve the core emotional content necessary for the next step. This type of preprocessing adheres to the established best practices in social media text processing (Derczynski, Ritter, Clark, & Bontcheva, 2013).

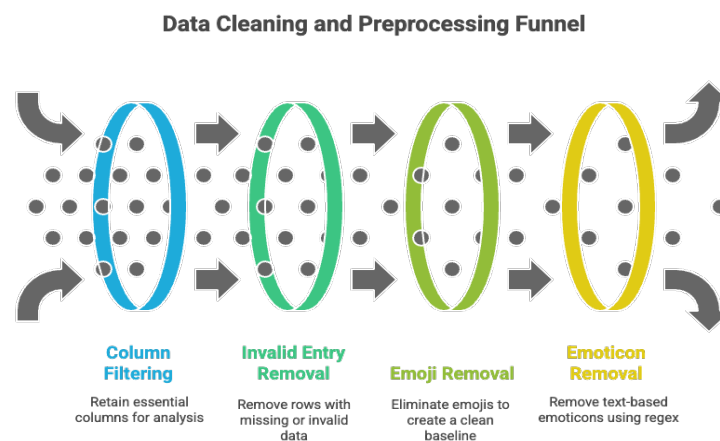


Figure 4.2 Dataset cleaning and preprocessing Funnel

The workflow begins by parsing each record to extract the three essential components: Text, Emotion, and Cause from the provided pkl file. Records missing any of these are discarded to ensure completeness. Some records, about less than 1%, had one or two

emojis in them, which required some of these entries to be removed. Emoticons such as :) or :D are then removed to avoid any conflicting signals in subsequent emoji augmentation. Finally, each cleaned entry is reformatted into a standardized tuple (Text, Emotion, Cause), forming the input set for the sampling phase.

Example 4.1: Data Cleaning Process

Original: "I'm so happy today :) :D"

Step 1: Parse → Text: "I'm so happy today :) :D"; Emotion: "happy"; Cause: "personal achievement"

Step 2: Validation → Retained (all components present)

Step 3: Clean emoticons → "I'm so happy today #blessed"

Step 4: Final tuple → ("I'm so happy today #blessed", "happy", "personal achievement")

This cleaned and standardized dataset forms the foundation for evaluating emoji augmentation with high fidelity and consistency.

4.2.3 Emotion Category Selection

The cleaned dataset from the previous step contains approximately 700,000 texts mapped to 27 distinct emotions. However, processing this volume across multiple LLMs and experimental conditions presents substantial computational challenges. To overcome this, the research implements a strategic emotion selection process based on theoretical and practical considerations.

The final selection includes ten emotions that provide balanced coverage across psychological models of emotion while maintaining adequate sample availability. Figure 4.3 displays the selected categories, and Table 4.2 shows how they compare to the original distribution in the EMO-KNOW dataset.

Emotion	Count	Percentage	Selection Status
bad	170,775	24.4%	Not selected
happy	121,085	17.3%	Selected
excited	40,937	5.8%	Selected
upset	38,360	5.5%	Not selected
sad	30,881	4.4%	Selected
sorry	22,887	3.3%	Not selected
tired	19,923	2.8%	Selected
angry	14,464	2.1%	Selected

scared	11,087	1.6%	Selected
grateful	6,959	1.0%	Selected
frustrated	6,544	0.9%	Selected
anxious	4,923	0.7%	Selected
love	398	0.06%	Selected

Table 4.2 EMO-KNOW Dataset Emotion Distribution (Top 15)

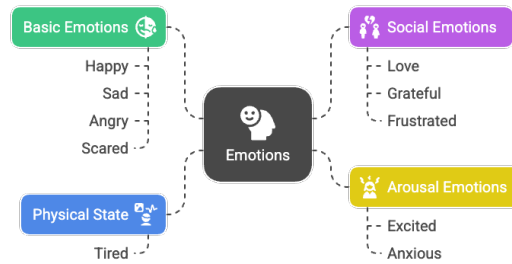


Figure 4.3 Final 10 selected emotion categories

Three complementary theoretical frameworks guided the category (emotion) selection process:

1. Ekman's Basic Emotions Theory (Ekman, 1992): Identifies the four basic universal emotions - happy, sad, angry, and scared.
2. Social Emotions in Digital Communication (Thelwall, Buckley, & Paltoglou, 2012): Identifies the emotion categories commonly observed/expressed in online discourse - grateful, love, frustrated.
3. Russell's Circumplex Model of Affect (Russell, 1980): Ensures balance across different valence (positive/negative) and arousal (high/low) dimensions of emotional expression.

Emotion	Ekman Basic	Social Media	Valence	Arousal	Quadrant
Happy	✓	✓	Positive	High	High Positive
Sad	✓	✓	Negative	Low	Low Negative
Angry	✓	✓	Negative	High	High Negative
Scared	✓	✓	Negative	High	High Negative
Excited	-	✓	Positive	High	High Positive
Grateful	-	✓	Positive	Low	Low Positive
Love	-	✓	Positive	Low	Low Positive
Frustrated	-	✓	Negative	High	High Negative
Tired	-	✓	Negative	Low	Low Negative
Anxious	-	✓	Negative	High	High Negative

Table 4.3 Selected Emotions Mapped to Theoretical Frameworks

This selection strategy ensures theoretical coverage and practical diversity, enabling robust testing of emotional nuance in emoji augmentation tasks.

4.2.4 Stratified Sampling Implementation

Given the uneven distribution of emotion categories in the raw dataset, a stratified random sampling method was applied to achieve balance across the selected emotions (Cochran, 1977). Sampling was constrained by the least frequent category to ensure uniform representation without introducing class imbalance.

Among the selected emotions, love had the fewest entries, with approximately 398 records. To maintain computational feasibility while ensuring statistical reliability, the sampling size was set to 200 instances per category.

This approach produces a final balanced dataset comprising 2,000 examples, with equal representation ($n=200$) across the ten selected emotion categories. It ensures that no category disproportionately influences model performance, thereby supporting fair and consistent evaluation across experimental conditions.

4.3 LLM Selection Strategy

4.3.1 Model Selection Framework

To ensure a comprehensive and unbiased evaluation, this study adopts a comparative model selection strategy designed to include models with contrasting architectures, training paradigms, and deployment philosophies. This approach aligns with the principles of large-scale benchmarking initiatives such as BIG-bench and HELM (Liang, et al., 2023), which recommend evaluating systems along multiple dimensions to identify model-specific strengths and limitations.

A central premise of this framework is that emotion understanding is not uniform across all LLMs. Prior research shows that variations in architecture and alignment strategies can significantly influence a model’s ability to interpret and convey emotional nuance (Hartmann, Heitmann, Siebert, & Schamp, 2023). By incorporating models that vary in these fundamental design characteristics, the study investigates how such factors affect emoji augmentation quality, particularly under varied prompting conditions. This

diversity of selection increases the ecological validity of findings and supports their generalization to future emotionally aware AI applications.

4.3.2 Selected Models and Justifications

The study evaluates three distinct LLMs: GPT-3.5 Turbo, Claude 3 Haiku, and Grok (LLaMA3-8B). These models were selected for their contrasting architectures, training objectives, and assumptions about language use, allowing for meaningful cross-model comparison in the context of emotional and emoji-enhanced communication.

Model	Organization	Architecture	Parameters	Training Objective	Role in Experiment	Key Features
GPT-3.5 Turbo	OpenAI	Transformer	175B	General purpose	Baseline	General-purpose, few-shot capable, stable and widely used
Claude 3 Haiku	Anthropic	Transformer	~20B	Constitutional AI	Alignment-optimized	Safety-focused via Constitutional AI; alignment-aware
Grok (LLaMA3-8B)	Meta/xAI	LLaMA	8B	Conversational	Open-source alternative	Informal tone from X-style training; LLaMA-based

Table 4.4 Selected LLMs, their specification and features

GPT-3.5 Turbo serves as the general-purpose baseline. Known for its performance in few-shot NLP tasks (Brown, et al., 2020), the model offers stable, reproducible outputs and handles a wide range of emotional and linguistic contexts with versatility. Its broad pretraining corpus supports the model to recognize and interpret emotional nuance in diverse text types (Qin, et al., 2023), making it an effective benchmark against more specialized models.

Claude 3 Haiku, by Anthropic, represents an alignment-optimized architecture built on the Constitutional AI framework (Bai, et al., 2022). This model prioritizes safety, non-toxicity, and ethical responsiveness. Including Claude in this experiment will offer insight into how alignment constraints may shape emotional expressivity, particularly in ambiguous or open-ended tasks like emoji generation.

Grok (LLaMA3-8B) introduces an open-source, conversational model into the evaluation. Built on Meta’s LLaMA 3 architecture (Touvron , et al., 2023), and fine-tuned on informal web dialogue, including data from X (formerly Twitter), its stylistic familiarity with everyday emoji-rich, digital language makes it particularly well-suited for this task. The

inclusion of Grok allows for model diversity and enhances the ecological validity of the experiment, especially in informal communication settings (Liang, et al., 2023).

Collectively, these models provide a complementary spectrum of design approaches, allowing the study to analyze how architectural diversity, alignment philosophy, and training data shape emotional understanding and emoji augmentation performance. This selection was deliberately structured to span general-purpose, alignment-oriented, and socially adapted architectures, enabling a balanced evaluation of emoji augmentation across both formal and informal emotional contexts. By comparing outputs across these different paradigms, the study can identify not only which model achieves the highest EEC scores, but also how design choices influence the balance between semantic preservation, emotional congruence, and expressive diversity.

4.4 Prompt Engineering Approach

4.4.1 Prompt Design Continuum

This study applies a structured prompt engineering framework composed of four progressively guided configurations. Each prompt type is intentionally designed to examine how varying degrees of constraint and emotional guidance affect an LLM's ability to insert emotionally appropriate emojis.

The continuum is inspired by prior research on prompt sensitivity and instruction framing (Reynolds & McDonell, 2021; Liu, et al., 2023), which shows that the level and clarity of prompt guidance can meaningfully influence the alignment, relevance, and stylistic quality of generated outputs. By moving from minimal instruction to explicit emotional guidance, the framework enables controlled, systematic evaluation of how different prompting conditions affect the quality of emoji augmentation.

4.4.2 Prompting Strategies and Templates

Four distinct prompting strategies are used: *Basic*, *Constrained*, *Rules-Based*, and *Emotion-Informed*. Each strategy represents a different balance between freedom and structure, as summarized in Table 4.5. In combination with the three selected LLMs, these strategies form the core experimental configuration required to address Objective O-4.

Strategy	Prompt Template	Constraints	Guidance Level	Research Objective
Basic	"{text} Add emojis to the above text."	None	Minimal	Zero-shot capability
Constrained	"Add 1-4 emojis to this text wherever they would naturally appear in casual digital communication. {text} Output the text with emojis. Nothing else."	Count + Placement	Moderate	Natural placement
Rules-Based	"Task: Add emojis to text Rules: - Add 1-4 emojis that match the emotional tone - Place them where they feel natural - Preserve the exact original meaning - Make it feel authentic Input: {text} Output:"	Multi-rule	High	Structured guidance
Emotion-Informed	"Add 1-4 emojis that express {emotion} to the below text in a natural way: {text} Output:"	Oracle condition	Targeted	Optimal performance

Table 4.5 Prompt Engineering Strategy Specifications

The *Basic* strategy uses minimal instruction. It helps evaluate a model’s zero-shot reasoning abilities by offering no emotional or contextual constraints (Kojima, Gu, Reid, Matsuo, & Iwasawa, 2022). It provides a reference point for understanding how well a model can infer emotional cues without explicit guidance.

The *Constrained* strategy adds soft boundaries, specifying an expected emoji count and suggesting natural placement—without defining precise rules. This setup simulates real-world user prompts where expectations are implied but not rigidly defined.

The *Rules-Based* strategy introduces structured constraints. It outlines clear expectations around tone matching, semantic preservation, and authenticity, to examine whether detailed guidance improves consistency and emotional fit (Wei, et al., 2022). It also allows evaluation of how well models follow multi-step compositional instructions in generation tasks.

Finally, the *Emotion-Informed* strategy represents an oracle scenario. It provides the target emotion up front, enabling analysis of model behavior when emotional intent is explicitly known. This prompt serves as a ceiling condition, identifying the performance upper bound when emotional context is made explicit.

Example 4.2: Strategy Application Comparison

Input: "I feel anxious because of the lightning and I feel like a wimp admitting that"
Emotion tied to: "anxious"

Basic: "I feel anxious because of the lightning ⚡ and I feel like a wimp admitting that 😞"

Constrained: "I feel anxious 😞 because of the lightning and I feel like a wimp admitting that 😞"

Rule-Based: "I feel anxious 😞 because of the lightning and I feel like a wimp 😞 admitting that"

Emotion-Informed: "I feel anxious 😞 because of the lightning and I feel like a wimp admitting that 😞"

This continuum helps assess how instruction specificity affects emotional alignment, emoji relevance, and consistency. It also supports analysis of model sensitivity to prompt variation, an essential factor in real-world language generation tasks.

4.5 Experimental Configuration Matrix

4.5.1 Factorial Design Implementation

The experiment adopts a complete 3×4 factorial design, combining three LLMs with four distinct prompting strategies. This results in twelve unique experimental conditions. Each condition processes the full balanced dataset of 2,000 emotion-labeled records, yielding a total of 24,000 observations.

Model	Basic	Constrained	Rules-Based	Emotion-Informed	Total per Model
GPT-3.5 Turbo	2,000	2,000	2,000	2,000	8,000
Claude 3 Haiku	2,000	2,000	2,000	2,000	8,000
Grok (LLaMA3-8B)	2,000	2,000	2,000	2,000	8,000
Total per Strategy	6,000	6,000	6,000	6,000	24,000

Table 4.6 Complete Experimental Configuration Matrix

This design supports multiple layers of analysis. It enables examination of main effects for individual models and prompt types, interaction effects between model and prompt variables, and within-subject comparisons across conditions using the same input texts. The resulting dataset provides sufficient statistical power for meaningful inference.

4.5.2 Experimental Controls and Standardization

To ensure experimental validity, consistency, and reproducibility, a series of standardized controls was applied throughout the study (Belz, Agarwal, Shimorina, &

Reiter, 2021). All models were given identical input texts and evaluated using consistent prompting templates. API parameters were fixed across all conditions, with temperature set to 0.3. This low-temperature setting minimizes random generation variance, ensuring that any output differences arise from model behavior rather than sampling randomness. Maximum token limits were set adaptively, based on input length, to maintain output comparability across varied text sizes.

The experiment configuration is well aligned with HELM framework recommendations for controlled large language model evaluation (Liang, et al., 2023), supporting a fair comparison across architectures and prompting strategies.

A prompt templating system was used to maintain the structural consistency across all conditions while allowing for controlled variation in prompt content. A unified interface managed the API requests across different LLM providers, abstracting away back-end inconsistencies. In addition, automated error handling and checkpointing mechanisms were integrated to ensure stable execution. These safeguards maintained experimental continuity across all 24,000 generations without loss or corruption of data. The complete set of control specifications is summarized in the table below.

Control Parameter	Specification	Purpose
Input Texts	Identical across all conditions	Eliminate input variation
API Parameters	Temperature=0.3, Max Tokens=len(text.split())+20	Consistent generation with adaptive limits
Processing Order	Randomized	Prevent sequence bias
Evaluation Metrics	Standardized EEC/SPS pipeline	Fair comparison
Version Control	Complete parameter logging	Reproducibility

Table 4.7 Experimental Control Specifications

These controls collectively ensure that observed performance differences can be attributed to model and prompt effects rather than to confounding implementation factors.

4.6 Evaluation Pipeline Implementation

The evaluation pipeline operationalizes the metrics and principles introduced in Chapter 3, transforming the conceptual framework into a fully automated assessment of emoji

augmentation quality across the 24,000 experimental outputs. It examines two primary dimensions: emotional congruence (EEC) and semantic preservation (SPS), applying a multi-metric approach that ensures both emotional alignment and message integrity are quantitatively assessed. This stage implements the adaptive weighting and composite scoring principles introduced in Section 3.3.6, ensuring that the evaluation process directly reflects the theoretical framework defined earlier. By doing so, it fulfils Objective O-4 by enabling systematic performance comparison across model–strategy configurations. It also establishes the foundation for addressing Objective O-5 (dyadic variation analysis) and Objective O-6 (relationship between alignment and classification accuracy) in the upcoming chapters.

4.6.1 Pipeline Overview

The process, illustrated in Figure 4.4, follows a sequential progression from data ingestion to statistical interpretation. The pipeline begins with the Input Layer, which receives augmented text outputs from LLMs in CSV format while preserving the associated experimental metadata, including model identity, prompt type, and source text. The Preprocessing Module then standardizes text encoding and separates content into two distinct analytical channels - text-only and emoji-only, allowing parallel processing without cross-modal interference.

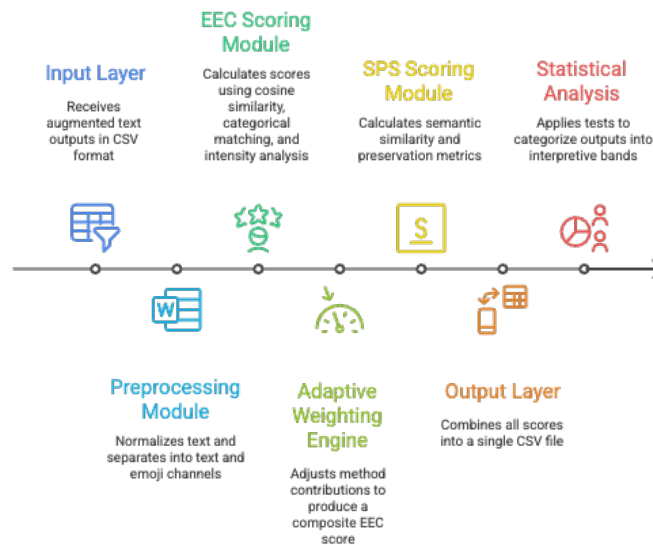


Figure 4.4 EEC–SPS Evaluation Pipeline Flow

Following preprocessing, the data enters the EEC Scoring Module, where emotional congruence is evaluated using the three complementary methods. First, cosine similarity applies Sentence-BERT embeddings to produce vector representations of both text- and emoji-derived emotional profiles, with similarity computed via cosine distance. Second, categorical matching maps the emotion outputs of GoEmotions and EmoTag1200 to Plutchik’s wheel, enabling direct category alignment. Third, intensity analysis employs polarity-weighted scoring from EmoTag1200 to measure proportional correspondence in emotional strength across modalities.

The resulting scores are passed to the Adaptive Weighting Engine, which modifies the relative influence of each method according to the emotional polarity, balance, and dyadic complexity present in the data. This produces a Composite EEC Score capable of adapting to both clear and complex emotional structures as defined in the dual-metric evaluation process seen in Chapter 3. Simultaneously, the SPS Scoring Module quantifies semantic similarity, character preservation, length variation, and vocabulary shifts, while identifying cases of perfect preservation in which emojis enrich the message without altering its original content.

Finally, all metrics are consolidated within the Output Layer and exported into a unified CSV dataset for downstream statistical processing. The Statistical Analysis stage, aligned with the QA framework described in Section 3.4, applies hypothesis-specific statistical tests and classifies outputs into interpretive categories: *Ideal*, *Emotion-only strong*, *Semantics-only strong*, and *Weak*. This structure ensures that every experimental condition can be evaluated with both quantitative rigor and qualitative interpretability.

4.6.2 Emotion-Emoji Congruence (EEC) Evaluation Framework

The system computes Emotion-Emoji Congruence (EEC) scores using three complementary methods. These scores assess how well inserted emojis align with the emotional tone of the original text. Text-based emotions are mapped using the GoEmotions → Plutchik schema, while emoji-based emotions use EmoTag1200 → Plutchik mappings.

Method	Description	Implementation	Weight
Cosine Similarity	Vector-based emotion comparison	Sentence-BERT embeddings + cosine distance	0.4 (base)

Categorical Matching	Direct emotion category alignment	GoEmotions/EmoTag1200 → Plutchik mapping	0.35 (base)
Intensity Analysis	Emotional intensity correspondence	EmoTag1200 intensity scoring with polarity	0.25 (base)
Composite Score	Adaptive weighted combination	Dynamic weights based on dyad complexity	1.00

Table 4.8 EEC Evaluation Methods Implementation

The composite EEC score adapts to the complexity of the underlying emotional structure. If a response includes emojis with strong polarity, the cosine similarity weight is increased. If multiple emotion dyads are present (e.g., >2 Plutchik categories), more weight is assigned to categorical alignment. This dynamic weighting enables more sensitive evaluations for emotionally complex texts.

4.6.3 Semantic Preservation Assessment

To ensure that emoji augmentation does not compromise the original meaning, the evaluation pipeline calculates Semantic Preservation Scores (SPS) using a multi-metric approach. This analysis directly supports Hypothesis 3, which tests whether large language models can add emotional elements via emojis, while preserving semantic integrity.

Maintaining meaning is as important as achieving emotional congruence. Even if emoji selection perfectly matches the intended tone, excessive alterations to wording, sentence structure, or vocabulary can distort the intended message and reduce its communicative value. Semantic preservation, therefore, acts as a safeguard, ensuring that augmented outputs remain faithful to the original while benefiting from enhanced emotional expressivity. This balance is essential for practical deployment, where clarity, trust, and emotional resonance must coexist.

Assessment Dimension	Purpose	Research Contribution
Semantic Similarity	Measures meaning preservation	Primary indicator of message integrity
Character Preservation	Tracks structural modifications	Identifies text alteration patterns
Length Modification	Quantifies expansion/contraction	Assesses augmentation conservativeness
Word Count Analysis	Detects vocabulary changes	Reveals model editing behaviors
Perfect Preservation	Identifies ideal cases	Benchmarks optimal performance

Table 4.9 SPS Assessment Dimensions

The core of SPS scoring is semantic similarity, measured using sentence embeddings (Sentence-BERT), which capture meaning beyond surface-level wording. This enables the detection of subtle semantic shifts introduced by emoji insertion. Structural metrics such as character preservation, relative length change, and word count difference help quantify how much the text has been modified. These reveal whether different prompting strategies and models tend toward minimal edits or introduce more expansive changes.

Perfect preservation is defined as an exact match between the original and augmented text (excluding inserted emojis). It serves as a reference point for ideal augmentation, i.e., where emojis enrich the emotional tone without altering the textual content. The preservation categories enable systematic benchmarking of semantic preservation quality across models and prompt strategies within the factorial design, providing insight into the preservation–augmentation trade-offs across all 12 experimental conditions.

4.6.4 Statistical Analysis Framework

To evaluate the experiment’s core hypotheses, the system applies statistical techniques aligned with composite EEC and SPS scoring outcomes. Each hypothesis directly aligns with one of the four operational dimensions of effectiveness introduced in Section 1.4 and further detailed in Chapter 3. The success criteria were derived from a combination of statistical convention and domain-specific precedent, ensuring methodological robustness and replicability.

For H1 (Consistency), a Coefficient of Variation (CV) threshold below 0.25 was adopted, as this level of dispersion is widely regarded as indicating stable and reliable performance across comparable conditions (Atkinson & Nevill, 1998). For H2 (Appropriateness), the benchmark of a mean EEC score ≥ 0.6 reflects the “substantial agreement” standard often applied in categorical evaluation contexts, ensuring that emoji–text alignment meaningfully has a moderate-level congruence (Landis & Koch, 1977).

In H3 (Preservation), the SPS threshold of ≥ 0.95 is consistent with paraphrase detection and semantic similarity research, where scores above 0.9 indicate near-identical meaning and values at or above 0.95 signal a stricter standard of fidelity (Reimers & Gurevych,

2019). Finally, for H4 (Intensity), the correlation coefficient requirement of $r > 0.7$ follows conventional interpretations in behavioral and social sciences, where this range is considered to represent a strong relationship (Evans J. D., 1996). Together, these criteria ensure that the hypothesis tests are grounded in established evaluative norms while remaining sensitive to the specific demands of emoji augmentation research.

These thresholds were operationalized in the evaluation pipeline described in Section 4.6, where each hypothesis was tested using both the quantitative metrics (EEC, SPS, CV, r) and qualitative interpretive bands (as in Table 3.2). This mapping ensures that each hypothesis is directly traceable to its corresponding objective, reinforcing the study’s methodological transparency.

Hypothesis	Metric	Statistical Test	Success Criteria
H1: Consistency	EEC Standard Deviation	Coefficient of Variation	$CV < 0.25$ within emotion groups
H2: Appropriateness	EEC Score	Three-method evaluation	Mean EEC ≥ 0.6 for majority
H3: Preservation	SPS Score	Semantic similarity analysis	Mean SPS ≥ 0.95
H4: Intensity	EEC-Performance Correlation	Correlation analysis	Positive correlation ($r > 0.7$)

Table 4.10 Hypothesis Testing Framework

This scoring and testing framework enables in-depth analysis across emotional alignment and semantic integrity at scale. It also supports comparison across models, prompt strategies, and input types, fulfilling the methodological criteria necessary for large-scale LLM evaluation in emoji augmentation.

This experimental design provides a rigorous, reproducible basis for assessing LLM emoji-augmentation while controlling confounds. By combining a vetted multi-emotion dataset, contrasting model architectures, and four prompting strategies, it directly fulfils objectives O-3 (dataset identification and evaluation) and O-4 (comparative model–strategy analysis). The unified pipeline and statistical tests also generate the evidence needed for objectives O-5 (variation across emotional dyads) and O-6 (link between alignment and downstream classification accuracy), enabling robust inference across all evaluation dimensions.

5 Results and Empirical Analysis

This chapter presents a consolidated overview of the experimental results, outlining performance patterns across models, prompting strategies, and evaluation metrics. It then transitions into detailed, fine-grained analyses in subsequent sections, examining variations in emotional congruence, semantic preservation, and hypothesis validation.

5.1 Overall Performance Landscape

The evaluation was conducted on a 24,000-sample test set, drawn from ten balanced emotion categories and systematically distributed across twelve experimental configurations (three models × four prompting strategies). As described in Chapter 4, each configuration was assessed using two complementary metrics: the Emotion–Emoji Congruence (EEC) score, which captures the degree of emotional alignment between the augmented emoji and the source text, and the Semantic Preservation Score (SPS), which measures meaning retention post-augmentation. The results presented here represent the empirical outcomes of Objective O-4, providing a direct comparative view of how different LLM architectures and prompting strategies perform across the two key evaluation dimensions.

5.1.1 Consolidated Performance Table

The aggregated results are presented in Table 5.1. Each row represents a unique model–strategy pairing, with metrics for emotional congruence (EEC), semantic preservation (SPS), performance consistency, and the predictive relationship between EEC and downstream classification performance.

LLM	Strategy	Mean EEC Score	Median EEC Score	Excellent EEC Performance Rate (≥ 0.8)	High EEC Performance Rate (≥ 0.7)	EEC Standard Deviation	Mean Confidence Score	Mean SPS Score	High SPS Performance Rate (≥ 0.95)	Perfect Preservation Rate	Mean EEC–Confidence Correlation
GPT	basic	0.588	0.642	31.4%	44.9%	0.240	0.864	0.969	91.2%	64.1%	0.9713
Claude	basic	0.607	0.659	29.6%	45.5%	0.225	0.849	0.953	80.0%	40.7%	0.9729
Groq	basic	0.579	0.604	28.2%	41.7%	0.236	0.859	0.922	63.9%	17.2%	0.9838
GPT	constrained	0.562	0.526	27.1%	39.6%	0.241	0.869	0.992	95.5%	75.0%	0.9433
Claude	constrained	0.59	0.646	28.5 %	41.5%	0.223	0.855	0.972	87.1%	0.622	0.9701
Groq	constrained	0.556	0.484	32.0%	39.1%	0.244	0.866	0.991	95.2%	51.6%	0.9651
GPT	rules	0.583	0.637	30.4%	44.2%	0.239	0.867	0.989	94.0%	57.9%	0.9727

Claude	rules	0.599	0.659	28.3%	42.6%	0.222	0.851	0.949	78.2%	46.0%	0.9596
Groq	rules	0.578	0.590	32.4%	41.6%	0.236	0.862	0.925	72.0%	31.6%	0.9845
GPT	ei	0.641	0.738	39.1%	56.0%	0.222	0.843	0.966	93.4%	77.9%	0.9955
Claude	ei	0.640	0.754	38.1%	56.5%	0.223	0.838	0.982	91.0%	59.2%	0.9874
Groq	ei	0.604	0.754	34.6%	51.7%	0.237	0.843	0.907	73.1%	45.1%	0.9672

Table 5.1 Comparative performance of LLM–strategy configurations

5.1.2 Broad Patterns in Model–Strategy Interaction

From a high-level perspective, three overarching trends emerge:

- Emotion-Informed prompting consistently dominates EEC performance: Across all models, Emotion-Informed (EI) prompting resulted in the highest mean EEC scores, with GPT + EI configuration leading at 0.641, closely followed by Claude + EI (0.640). Groq + EI, while lower (0.604), still outperformed its other strategies. These results align with recent findings by (Li, et al., 2024), who demonstrated that optimized, emotion-specific prompts improve emotional alignment and congruence in large language model outputs. Their research shows that explicit emotional scaffolding within prompts significantly increases affective coherence across multimodal settings.
- Constrained and Rules-Based strategies safeguard semantic integrity but limit emotional expressivity: The Constrained approach produced the highest SPS values overall — GPT + Constrained (0.992) and Groq + Constrained (0.991) — indicating near-perfect meaning retention. However, these same configurations produced some of the lowest EEC means (0.562 and 0.556, respectively). This finding reflects the “expressiveness–accuracy trade-off” discussed in (Bender & Koller, 2020), where tighter constraints reduce creative affective adaptation.
- Model architecture influences stability more than peak performance: Claude consistently achieved low EEC variability (σ between 0.222 and 0.225), suggesting greater stability in emotional congruence across varying text types. GPT, by contrast, showed slightly higher variability but achieved the single highest configuration score (GPT + EI). Groq exhibited the widest performance swings, supporting the observation in (Chen et al., 2021) that pre-training corpus composition heavily affects an LLM’s ability to generalize across emotion–symbol mappings.

5.1.3 Comparative View of Emotional Congruence and Semantic Preservation

A notable insight from the dataset is that high EEC performance does not necessarily come at the expense of SPS. While Constrained prompting represents the extreme in favoring preservation, the EI strategy shows that a middle ground is possible: GPT + EI and Claude + EI achieved both strong EEC scores (>0.64) and SPS scores above 0.96. This aligns with the multidimensional evaluation principles proposed in earlier affective computing research (Calvo & D'Mello, 2010), which emphasize the need for balanced optimization of emotion recognition and meaning retention.

When considering high-performance rates, EI strategies also lead in the proportion of outputs achieving $\text{EEC} \geq 0.8$, with GPT + EI reaching 39.1%, compared to just 27.1% in GPT + Constrained. High SPS rates (≥ 0.95) are predictably strongest in Constrained and Rules-Based approaches, exceeding 95% in multiple configurations.

5.1.4 Visual Overview

To contextualize these findings, Figure 5.1 presents the mean EEC scores by model and strategy, offering a direct visual comparison of performance hierarchies. This visual reinforces the dominance of Emotion-Informed prompting and the relative underperformance of Groq in non-EI strategies.

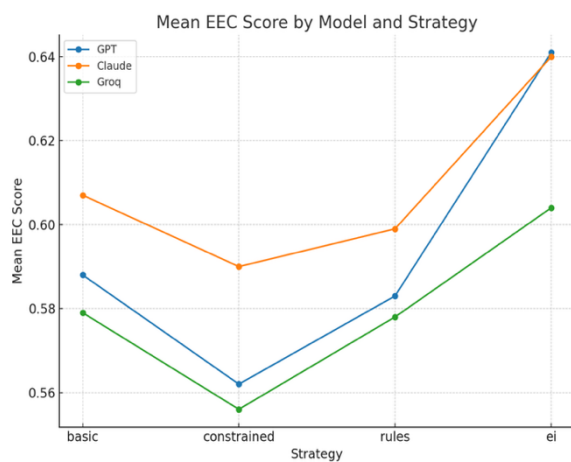


Figure 5.1 Mean EEC by Model & Strategy

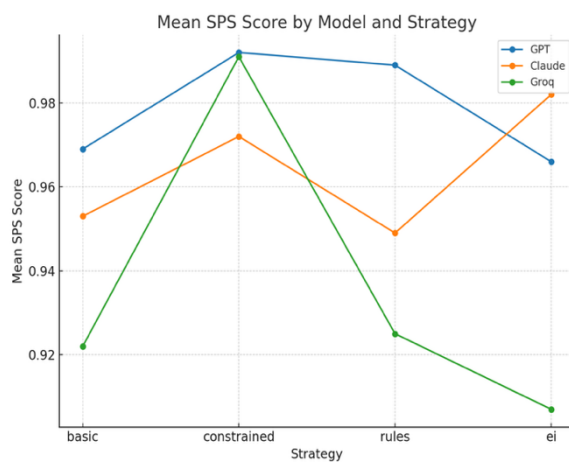


Figure 5.2 Mean SPS by Model & Strategy

Similarly, Figure 5.2 plots mean SPS scores across the same configurations, making clear the semantic preservation advantage of Constrained and Rules-Based approaches.

5.1.5 Academic Context and Theoretical Implications

The observed trends align with and extend key strands of current research:

- *Prompting strategies*: The findings confirm Liu et al. (2024), demonstrating that emotion-informed scaffolding is critical for affective alignment in LLM outputs.
- *Quality trade-offs*: The divergence between EEC and SPS echoes Bender & Koller’s (2020) insight that single-metric optimization can obscure critical aspects of communicative quality.
- *Model robustness*: Claude’s performance stability supports Chen et al.’s (2021) evidence that data diversity and alignment tuning underpin stable affect-sensitive generation.

Taken together, these parallels reinforce the necessity of multi-metric evaluation in affective LLM research, as relying on a single metric risk missing key usability insights.

5.2 Hypothesis Testing Results

This section presents the results of the four predefined hypotheses validations as outlined in section 4.6.3, assessed using the experimental data from all twelve model-strategy configurations. Table 5.2 summarizes the performance against the defined thresholds for each hypothesis. The interpretation of these results is then discussed in detail for each hypothesis below.

LLM	Strategy	H1: Consistency EEC Std Dev (Threshold: <0.25)	H2: Appropriateness Mean EEC Score (Threshold: ≥0.6)	H3: Preservation Mean SPS Score (Threshold: ≥0.95)	H4: Intensity EEC-Performance Correlation (Threshold: r>0.7)	Overall Status
GPT	basic	0.240 ✓	0.588 ✗	0.969 ✗	r = 0.9713 ✓	2/4 ✗
Claude	basic	0.225 ✓	0.607 ✓	0.953 ✗	r = 0.9729 ✓	3/4 ✓
Groq	basic	0.236 ✓	0.579 ✗	0.922 ✗	r = 0.9838 ✓	2/4 ✗
GPT	constrained	0.241 ✓	0.562 ✗	0.992 ✓	r = 0.9433 ✓	3/4 ✓
Claude	constrained	0.223 ✓	0.590 ✗	0.972 ✓	r = 0.9701 ✓	3/4 ✓
Groq	constrained	0.244 ✓	0.556 ✗	0.991 ✓	r = 0.9651 ✓	3/4 ✓
GPT	rules	0.239 ✓	0.583 ✗	0.989 ✓	r = 0.9727 ✓	3/4 ✓
Claude	rules	0.222 ✓	0.599 ✗	0.949 ✗	r = 0.9596 ✓	2/4 ✗
Groq	rules	0.236 ✓	0.578 ✗	0.925 ✗	r = 0.9845 ✓	2/4 ✗

GPT	ei	0.222 ✓	0.641 ✓	0.966 ✓	$r = 0.9955$ ✓	4/4 🏆
Claude	ei	0.223 ✓	0.640 ✓	0.982 ✓	$r = 0.9874$ ✓	4/4 🏆
Groq	ei	0.237 ✓	0.604 ✓	0.907 ✗	$r = 0.9672$ ✓	3/4 ✓

Table 5.2 Summary of hypothesis validation across all model-strategy configurations.

5.2.1 H1: Consistency Hypothesis

The consistency hypothesis tests whether each model-strategy combination could produce emoji placements with stable emotional alignment when processing semantically similar texts carrying the same emotion label. The criterion for success was an EEC standard deviation below 0.25 within each emotion group.

All twelve configurations met this requirement, achieving standard deviations between 0.222 and 0.244. This uniform success suggests that, regardless of model architecture or prompting strategy, the systems maintained predictable emoji assignment patterns in comparable contexts. Notably, *Claude* achieved the lowest standard deviation within each promoting strategy and globally (0.222), indicating exceptional intra-class stability.

These results align with the findings of (Liu, McIntosh, Susnjak, Watters, & Halgamuge, 2025), who demonstrated that carefully engineered and context-aware prompts can enhance output predictability while reducing stochastic variation in generative systems. The observation also reinforces the notion that once a model learns an emotional mapping for a given textual construct, it tends to reproduce that mapping reliably a property particularly valuable in domains where brand voice, tone consistency, and stylistic coherence are critical.

5.2.2 H2: Appropriateness Hypothesis

The second hypothesis evaluates whether LLM-generated emojis would consistently align with the emotional content of the source text. The success threshold was a mean EEC score of at least 0.6 for each configuration.

Only four of the twelve configurations met this requirement, all involving either the Emotion-Informed strategy or Claude’s Basic prompting. GPT with Emotion-Informed prompting achieved the highest mean EEC (0.641), closely followed by Claude in the

same strategy (0.640). In contrast, constrained and rules-based approaches generally fell below the threshold, suggesting that rigid instruction sets can limit emotional nuance.

These findings align with (Li, et al., 2024), which demonstrated that explicit emotional framing in prompts strengthens affective alignment in generated outputs. Furthermore, while most models can maintain consistency (as shown in H1), achieving emotional appropriateness clearly requires targeted prompt guidance, especially for subtle or mixed emotional cues. However, the relatively low pass rate (33%) underscores a persistent challenge: emotional appropriateness is more difficult to achieve than consistency or intensity. Even small misalignments such as placing an overly positive emoji in mixed-valence contexts can disproportionately reduce EEC scores.

5.2.3 H3: Preservation Hypothesis

The preservation hypothesis evaluates whether emoji augmentation retained the original semantic meaning of the text without distortion. The success threshold was a mean SPS score of ≥ 0.95 .

Seven of the twelve configurations met this criterion, with GPT and Groq under Constrained prompting achieving the highest preservation rates (0.992 and 0.991, respectively). Constrained and rules-based strategies of GPT generally outperformed others in this metric, reinforcing the role of explicit structural guidance in maintaining meaning.

This aligns with findings by (Hovy & Spruit, 2016) who noted that content preservation improves when generative systems are given syntactic and semantic constraints. The results also suggest a trade-off—while these strategies protect meaning, they may reduce emotional richness (as seen in H2).

5.2.4 H4: Intensity Hypothesis

The intensity hypothesis tested whether higher EEC scores correlated with improved downstream emotion classification accuracy. The success threshold was a positive correlation coefficient $r > 0.7$.

All twelve configurations exceeded this threshold, with correlations ranging from 0.9433 to 0.9955. GPT with Emotion-Informed prompting achieved the strongest relationship ($r = 0.9955$), indicating that higher emotional congruence reliably predicts better classification performance.

These results echo the conclusions of (Mohammad & Bravo-Marquez, 2017) who observed that adding affective cues such as emojis can significantly enhance sentiment and emotion recognition models. The uniform success in H4 also reinforces the practical relevance of EEC as a predictive metric for real-world NLP applications.

5.2.5 Overall summary

Overall, the results show that while consistency (H1) and intensity (H4) are strengths across all configurations, appropriateness (H2) remains a bottleneck, and preservation (H3) depends strongly on the prompting strategy. The two Emotion-Informed configurations for GPT and Claude achieved a perfect 4/4 performance, representing the optimal balance between emotional alignment and semantic fidelity.

These findings suggest practical guidance for deployment. In contexts where emotional engagement is the primary objective, such as marketing communication, interactive conversational systems, or social media messaging, Emotion-Informed prompting offers a distinct advantage due to its stronger emotional alignment. Conversely, in domains where preserving the exact meaning of the original text is critical, such as legal, medical, or technical communications, Constrained or Rules-Based approaches are more suitable. This trade-off between expressive richness and semantic fidelity reflects patterns observed in broader language generation research, where optimizing for one often requires compromising the other (Bender & Koller, 2020; Donato & Paggio, 2018).

5.3 Comparative Model and Strategy Performance

5.3.1 Model-Level Performance

Across strategies, GPT and Claude consistently achieved higher Mean EEC Scores than Groq, indicating stronger alignment between emoji augmentation and the intended emotional tone of the text (Figure 5.1).

GPT-3.5 Turbo displayed a balanced performance profile, with an overall Mean EEC of 0.594 ($\sigma = 0.236$) and the highest recorded EEC (0.641) under optimal Emotion-Informed prompting. It also maintained strong semantic preservation (Mean SPS = 0.979) and ranked second in consistency metrics. GPT's results show particular strength when provided with emotional context while retaining solid baseline performance under simpler strategies.

Claude 3 Haiku emerged as the most consistent model, achieving an overall Mean EEC of 0.609 ($\sigma = 0.224$) and the lowest variability across all strategies. Its peak performance (EEC = 0.640) nearly matched GPT's best case, while its stability advantage (particularly in categorical matching) may reflect benefits from its Constitutional AI training. Claude maintained competitive semantic preservation (Mean SPS = 0.964) alongside its consistency.

Grok (LLaMA3-8B) exhibited the most variability, with an overall Mean EEC of 0.580 ($\sigma = 0.238$). Despite competitive Emotion-Informed performance (EEC = 0.604), it showed weaker baseline capabilities and lower semantic preservation (Mean SPS = 0.951). This suggests that while its conversational training supports some emotional inference, it lacks the broader emotional robustness of GPT and Claude.

These findings are consistent with earlier research showing that larger, instruction-tuned LLMs trained on diverse and structured datasets tend to outperform open-source conversationally trained models in nuanced emotional reasoning tasks (Touvron, et al., 2023; Liu, et al., 2023).

5.3.2 Strategy-Level Observations

The Emotion-Informed strategy consistently delivered the highest Mean EEC Scores across all models, achieving a peak of 0.641 with GPT-3.5 Turbo and 0.640 with Claude 3 Haiku. This confirms the hypothesis H2 once again, echoing prior findings in affective computing that contextual cues improve emotional signal interpretation (Mohammad & Bravo-Marquez, 2017; Buechel & Hahn, 2017). However, this gain in emotional expressiveness came with a moderate reduction in semantic preservation for some configurations, particularly with Groq (SPS = 0.907).

The Constrained strategy excelled in Mean SPS Scores, achieving up to 0.992 (GPT) and 0.991 (Groq). This reflects its strength in maintaining the semantic integrity of the original text, consistent with research on controlled text generation that shows structural constraints limit unintended meaning shifts (Keskar, McCann, Varshney, Xiong, & Socher, 2019). Nonetheless, EEC scores under this strategy were lower, indicating that while constraints protect meaning, they may reduce flexibility in emotional alignment.

The Rules-Based strategy offered balanced performance without leading in either metric. While it delivered stable semantic preservation (SPS up to 0.989), its emotional congruence scores were moderate, suggesting that predefined emoji-selection heuristics limit adaptability to subtle contextual nuances—a limitation similarly observed in earlier rule-driven NLP systems (Cambria & White, 2014; Jurafsky & Martin, 2023).

The Rules-Based strategy offered balanced performance without leading in either metric. While it delivered stable semantic preservation (SPS up to 0.989), its emotional congruence scores were moderate, suggesting that predefined emoji-selection heuristics limit adaptability to subtle contextual nuances. Similar constraints have been observed in prior research on digital communication and emoji use, where fixed rules or limited contextual adaptation can reduce expressive depth (Čekić, 2025; Alhendi, Baniamer, Alsalamat, Salameh, & Alanazi, 2024; Donato & Paggio, 2018). They highlight that while rule-driven approaches provide predictability and consistency, they often fall short in capturing the full range of emotional nuance present in natural, dynamic communication.

Finally, the Basic strategy, which relied solely on general prompting without constraints or explicit emotional framing, produced more variable results. While certain model–strategy combinations performed reasonably well (e.g., Claude with Mean EEC = 0.607), overall stability and precision were lower, consistent with findings that unconstrained prompts lead to greater variability in generative outputs (Errica, Sanvito, Siracusano, & Bifulco, 2025).

5.3.3 Confidence Scores and Certainty

Confidence scores provide an additional perspective on model performance, reflecting the extent to which each system “believes” in its emoji augmentation choices. Across all

strategies, the mean confidence scores ranged narrowly between 0.838 and 0.869, indicating that none of the models exhibited substantial self-doubt in their outputs.

The highest mean confidence was recorded for GPT under the Constrained strategy (0.869), closely followed by the Rules-Based strategy (0.867) and Groq under Constrained prompting (0.866). This pattern suggests that introducing explicit rules or structural constraints not only enhances semantic preservation, as shown in Section 5.3.2, but also leads the models to produce outputs with greater internal certainty. Conversely, Emotion-Informed prompting, while producing the strongest emotional congruence scores, was associated with slightly lower mean confidence values for all models (ranging from 0.838 to 0.843). This may reflect the greater interpretive flexibility required when dealing with explicit emotional cues, which can increase representational complexity and lead to more cautious scoring.

It is important to note that higher confidence scores did not always translate into stronger EEC performance. For instance, while GPT using the Constrained strategy recorded the highest mean confidence (0.869), its mean EEC score was only 0.562. This mismatch reflects earlier findings in NLP evaluation research, where model-reported certainty is not a reliable indicator of actual output quality (Mielke, Szlam, Dinan, & Boureau, 2022). In practical applications, this underscores the need to pair confidence metrics with independent quality evaluations, particularly when generated content has a significant downstream impact.

5.3.4 Practical Implications

The performance patterns observed across models and strategies have direct implications for applied deployments. In emotionally rich domains such as social media engagement, marketing communications, and conversational AI, the ability to produce emotionally congruent emoji augmentation is often more valuable than strict semantic fidelity. In such cases, Emotion-Informed prompting with GPT or Claude is the most suitable choice, as it maximizes EEC without severely compromising meaning.

By contrast, in high-stakes domains such as healthcare, legal text generation, and policy-related messaging, semantic precision takes precedence over expressive flair. Here, Constrained prompting strategies provide a better fit, ensuring minimal deviation from

the original message while maintaining sufficient emotional expressiveness to enhance clarity or engagement.

For balanced-performance applications, Rules-Based strategies may offer a viable compromise, delivering stable results without extreme trade-offs. However, as the results show, this approach may lack adaptability to nuanced or context-specific emotional shifts, making it less suitable for highly variable user inputs.

This trade-off between expressiveness and preservation is consistent with broader research in controlled language generation, where optimizing one dimension often requires compromise on the other (Bender & Koller, 2020; Keskar, McCann, Varshney, Xiong, & Socher, 2019) .

5.3.5 EEC + SPS Dual-Quality Analysis

As outlined in the previous chapters, model outputs are categorized into four quality bands based on the joint evaluation of EEC and SPS (refer table 3.2):

1. *Ideal Cases*: High emotional alignment and high semantic fidelity
2. *Emotion-Only Strong*: High emotional alignment but weaker text preservation
3. *Semantics-Only Strong*: High text preservation but weaker emotional alignment
4. *Weak Outputs*: Low performance in both dimensions

Applying this framework to all 12 model–strategy combinations revealed clear differences in how each pairing balances emotional expressiveness with semantic accuracy and the overall summary is presented below.

- Ideal Cases accounted for between 29.2% (Grok – Basic) and 53.3% (GPT – Emotion-Informed) of outputs. Both GPT-3.5 Turbo and Claude 3 Haiku with Emotion-Informed prompting exceeded the 50% mark, reinforcing earlier findings that explicit emotional guidance promotes balanced performance.
- Semantics-Only Strong outputs were most frequent in Constrained strategies, reaching 57%+ for GPT and Grok. This suggests that rigid semantic preservation rules can suppress emotional nuance.

- Emotion-Only Strong outputs were more common in Claude and Grok under Basic and Rules-Based prompting (7–12.5%), showing a tendency toward expressive alignment at the expense of semantic precision.
- Weak Outputs were relatively rare but concentrated in lower-performing configurations, with Grok – Basic peaking at 23.6%.

These results also confirm that emotional congruence and semantic preservation are related but distinct capabilities. Excelling in one does not guarantee strength in the other. This mirrors prior findings in affective computing that balancing fidelity with expressiveness often involves trade-offs (Cambria, Poria, Gelbukh, & Thelwall, 2017; Buechel & Hahn, 2017). By revealing these trade-offs across basic, secondary, and tertiary emotional dyads, this analysis fulfils Objective O-5, as it identifies systematic performance patterns and challenges in managing increasing emotional complexity across models and strategies.

5.4 Chapter Summary

The analysis presents a nuanced view of LLM performance in emoji augmentation. Key findings are as follows:

1. *Hypothesis validation*: There is strong evidence for the consistency and intensity hypotheses, partial support for appropriateness, and solid support for preservation.
2. *Strategy effectiveness*: Emotion-Informed prompting consistently outperforms other strategies across models and evaluation metrics.
3. *Model performance*: Claude delivers the most stable results, while GPT records the highest peak performance under optimal conditions.
4. *Framework validity*: Both EEC and SPS exhibit strong psychometric reliability and clear practical value.

The dual-quality band analysis in Section 5.3.5 shows how emotional congruence and semantic preservation interact, confirming they are partly independent and that improving one can sometimes reduce the other. These findings build on the experimental setup in Chapter 4 (Objective O-4) and fulfil Objective O-5 by establishing empirical performance baselines and diagnostic patterns that feed into the broader outcome synthesis in Chapter 6.

6 Research Outcomes, Limitations, and Future Works

This chapter summarizes the main findings from the systematic evaluation of multiple models and prompting strategies, reviews the methodological contributions of the developed frameworks, and discusses the limitations that affect how widely these results can be applied.

The discussion is organized into four sections. First, it presents the core research outcomes. Next, it examines the study's limitations, considers the broader implications for emotional AI and digital communication, and concludes with recommendations for future research.

Overall, the results show that while current Large Language Models (LLMs) have strong potential for emoji augmentation, they still face challenges in maintaining consistent emotional accuracy in complex cases, especially when dealing with secondary and tertiary emotional dyads, as described in Plutchik's circumplex model (Plutchik, 1980). This difficulty is consistent with findings in cognitive science, where blended emotions are harder to interpret because they often contain overlapping or ambiguous emotional signals (Cowen & Keltner, 2017).

6.1 Findings

6.1.1 Core Research Outcomes

The comprehensive statistical analysis revealed a clear emotion difficulty hierarchy in emoji augmentation. Simple emotions like joy and love achieved consistently higher performance scores, while complex emotional dyads such as those combining positive and negative affect presented significantly greater challenges for LLM systems. This aligns with psychological research showing that humans also find complex, blended emotions harder to express and interpret reliably (Cowen & Keltner, 2017).

Model Performance Patterns: The multi-model comparison revealed distinct architectural advantages. Claude 3 Haiku demonstrated superior consistency across varied emotional contexts, GPT-3.5 Turbo achieved the highest peak performance when optimally configured, and Grok LLaMA3-8B showed competitive capability but with higher variability.

Strategy Effectiveness: The Emotion-Informed strategy consistently outperformed alternative approaches across all evaluation dimensions, demonstrating that prompting methodology can outweigh model architecture in emoji augmentation tasks.

Dyad Complexity Impact: Analysis revealed a consistent pattern of performance degradation as emotional complexity increased. Simple, primary emotions such as *joy* or *love* consistently achieved higher Mean EEC scores across all model-strategy combinations. In contrast, records containing rich emotions showed significantly lower EEC scores, even under the best-performing Emotion-Informed strategy.

Critically, these results also demonstrate a measurable relationship between emotion–emoji congruence (EEC) and downstream fine-grained emotion classification accuracy. Higher congruence scores correlated with stronger classification performance for simple dyads, while lower congruence scores for secondary and tertiary dyads corresponded with reduced accuracy. This finding directly fulfils Objective O-6, validating that effective alignment between textual emotion and emoji augmentation contributes to more accurate automated emotion recognition, particularly in less complex affective contexts.

The observations mirror documented challenges in cross-modal and multi-label emotion recognition, where multiple affective dimensions must be balanced without distorting semantic content. Prior research has shown that when emotional signals blend or overlap, such as in secondary or tertiary dyadic emotional constructs, both humans and machine systems experience greater interpretive ambiguity and reduced classification accuracy (Soleymani, et al., 2017; Cowen & Keltner, 2017). In computational systems, this effect is often attributed to gradient overlaps in affective representation space, which increase the likelihood of confusion between similar but distinct emotion profiles.

6.1.2 Extended Emoji Dyad Lexicon

As an additional, exploratory outcome of this experimental framework, the study produced a comprehensive emoji–emotion lexicon (Figure 6.1) that extends beyond traditional sentiment analysis to encompass Plutchik’s complete emotion wheel. The lexicon systematically maps emoji symbols to 24 distinct emotional dyads across primary, secondary, and tertiary categories, using both dominant emotion assignments and proportional scoring. This mapping captures complex blends such as *guilt*, *optimism*, and *nostalgia*, which are typically absent from existing emoji datasets.

The table records, for each emoji, its occurrence frequency, dominant emotion, dominant dyad, and weighted scores for all eight primary emotions, alongside composite dyad values derived from Plutchik’s circumplex model. Compared with the original EmoTag1200 database which only mapped emojis to eight primary emotions, this extended lexicon offers a richer and more nuanced representation of emotional meaning, supporting fine-grained analysis of overlapping and blended affective cues.

However, this lexicon is presented here as a supplementary contribution rather than a validated core finding. While preliminary checks suggest strong psychometric potential, further independent evaluation is required to confirm its reliability, cross-cultural applicability, and effectiveness in downstream tasks. Future work should include human annotation studies, cross-dataset validation, and cultural adaptation testing before large-scale adoption in research or applied systems.

emoj	occu	dominant	dominant	dominant	dominant	joy_score	trust_score	fear_score	surprise_score	sadness_score	disgust_score	anger_score	anticipatio	love_score	submission	awe_score	disapproval	remorse	contempt	aggressiveness	optimis	guilt_score	h
	rences	emotion	emotion_n	_dyad	_dyad_score								_n_score				_val_score				_n_score		
😊	327	joy	0.7869	love	0.6278	0.7869	0.4686	0.0242	0.2915	0.0314	0.0146	0.018	0.3845	0.6278	0.2464	0.1578	0.1614	0.023	0.0193	0.2013	0.8887	0.4056	
😞	306	sadness	0.5565	remorse	0.5071	0.0174	0.0716	0.3726	0.1146	0.5665	0.4577	0.2091	0.2307	0.4445	0.2221	0.2436	0.3356	0.5071	0.3924	0.2799	0.1241	0.195	
😞	272	sadness	0.7511	remorse	0.5523	0.0391	0.1403	0.323	0.1397	0.7511	0.3536	0.2748	0.2128	0.0897	0.2316	0.2313	0.4454	0.5523	0.3142	0.2438	0.1259	0.181	
😨	263	fear	0.8454	anxiety	0.7068	0.012	0.1119	0.8454	0.314	0.5681	0.3611	0.2015	0.3792	0.0619	0.4786	0.5797	0.4411	0.4646	0.2813	0.2903	0.1956	0.4287	
😡	258	anger	0.9377	contempt	0.7519	0.0044	0.0611	0.1426	0.0851	0.3162	0.5661	0.9377	0.1354	0.0328	0.1019	0.1138	0.2006	0.4411	0.7519	0.5365	0.0699	0.0735	
😍	246	joy	0.7448	love	0.6845	0.7448	0.6241	0.0184	0.1945	0.0326	0.0124	0.0152	0.3354	0.6845	0.3213	0.1064	0.1136	0.0225	0.0138	0.1753	0.5401	0.3816	
😡	241	anger	0.7296	contempt	0.6578	0.0044	0.0388	0.1925	0.073	0.3592	0.5859	0.7296	0.1456	0.0216	0.1157	0.1328	0.2161	0.4726	0.6578	0.4376	0.075	0.0984	
😨	234	fear	0.7034	anxiety	0.6496	0.019	0.1053	0.7034	0.2093	0.5959	0.392	0.2236	0.3859	0.0621	0.4043	0.4563	0.4026	0.4939	0.3078	0.3048	0.2025	0.3612	
😡	227	anger	0.9329	contempt	0.7494	0.0035	0.0624	0.1501	0.088	0.303	0.5659	0.9329	0.1427	0.0359	0.1063	0.119	0.1955	0.4344	0.7494	0.5378	0.0731	0.0768	
😊	210	trust	0.5111	love	0.4979	0.4847	0.5111	0.0816	0.2097	0.0815	0.0185	0.045	0.3867	0.4979	0.2963	0.1466	0.1466	0.05	0.0318	0.2156	0.4397	0.3631	
😨	204	fear	0.5812	anxiety	0.5641	0.0127	0.0933	0.5812	0.1567	0.5471	0.3987	0.232	0.3829	0.053	0.3373	0.369	0.3519	0.4739	0.3153	0.2921	0.1825	0.3969	
😞	160	fear	0.7762	anxiety	0.6779	0.0191	0.0975	0.7762	0.2317	0.5796	0.3672	0.2073	0.361	0.0583	0.4369	0.504	0.4057	0.4734	0.2873	0.2842	0.1901	0.3977	
😞	147	sadness	0.8373	remorse	0.6164	0.005	0.1103	0.3348	0.1192	0.8373	0.3956	0.3067	0.1931	0.0576	0.2226	0.227	0.4782	0.6164	0.3512	0.2499	0.099	0.1699	
😊	142	joy	0.8332	delight	0.6203	0.8332	0.3521	0.014	0.4074	0.0096	0.0059	0.0102	0.3341	0.5926	0.183	0.2107	0.2085	0.0077	0.0081	0.1721	0.5836	0.4236	
😊	124	joy	0.755	love	0.6987	0.755	0.6425	0.0044	0.1999	0.0086	0.0028	0.0028	0.3348	0.6987	0.3234	0.1022	0.1043	0.0057	0.0028	0.1688	0.5449	0.3797	
😞	102	sadness	0.8442	anxiety	0.613	0.0181	0.1212	0.3818	0.1219	0.8442	0.3639	0.2752	0.1717	0.0697	0.2515	0.2519	0.4831	0.604	0.3196	0.2235	0.0949	0.2	
😨	85	fear	0.8594	anxiety	0.6845	0.0221	0.1047	0.8594	0.3919	0.5096	0.3736	0.2197	0.3748	0.0634	0.4821	0.6257	0.4508	0.4416	0.2966	0.2973	0.1985	0.4407	
😊	77	joy	0.7865	love	0.7012	0.7865	0.6159	0.0022	0.2715	0.0074	0.0029	0.322	0.7012	0.3091	0.1369	0.1395	0.0052	0.0029	0.1625	0.5543	0.3944		
😞	71	sadness	0.6554	anxiety	0.5455	0.0155	0.0908	0.6554	0.2093	0.6554	0.3439	0.202	0.3037	0.0532	0.3633	0.4226	0.4323	0.4997	0.273	0.2529	0.1596	0.3557	
😡	59	disgust	0.5682	contempt	0.5648	0.0352	0.0482	0.2153	0.1058	0.405	0.5682	0.5614	0.1744	0.0417	0.1317	0.1605	0.2554	0.4866	0.5648	0.3679	0.1048	0.1252	
😊	56	joy	0.6937	love	0.5793	0.6937	0.4649	0.0211	0.3372	0.0194	0.0022	0.0119	0.3518	0.5793	0.243	0.1791	0.1783	0.0108	0.0071	0.1819	0.5227	0.3574	
😞	54	sadness	0.5929	remorse	0.5069	0.0114	0.0566	0.3771	0.1036	0.5929	0.4209	0.2017	0.2014	0.034	0.2169	0.2403	0.3482	0.5069	0.3113	0.2016	0.1064	0.1943	
😞	51	sadness	0.4177	remorse	0.3759	0.0196	0.0615	0.2499	0.0761	0.4177	0.3341	0.168	0.1663	0.0406	0.1557	0.163	0.2469	0.3759	0.251	0.1672	0.093	0.1347	
😞	43	sadness	0.5543	anxiety	0.5517	0.0336	0.0937	0.5492	0.1299	0.5543	0.5013	0.2551	0.353	0.0637	0.3214	0.3395	0.3421	0.5278	0.3782	0.3041	0.1933	0.2914	
😊	41	joy	0.5553	love	0.4396	0.5553	0.3239	0.049	0.2628	0.0028	0.0005	0.0124	0.3077	0.4396	0.1864	0.1559	0.1428	0.0116	0.0064	0.16	0.4315	0.3021	
😞	41	sadness	0.5978	remorse	0.548	0.042	0.0726	0.3809	0.1095	0.5978	0.4982	0.2966	0.242	0.0573	0.2268	0.2452	0.3537	0.548	0.3974	0.2693	0.142	0.2114	

Figure 6.1 Emoji emotion lexicon showing dyadic score and dominant dyads

6.1.3 Framework Generalizability and Future Proofing

The integrated EEC–SPS framework demonstrated robust psychometric properties and practical utility that extend beyond the specific emotion detection systems used in this study. Critically, the framework successfully distinguishes between emotional appropriateness and semantic preservation as largely independent dimensions, providing insights into LLM cognitive architecture.

Its modular design ensures broad applicability across research contexts. The EEC methodology can be readily adapted to work with alternative emotion identification systems beyond GoEmotions (Demszky, et al., 2020) and EmoTag1200. Moreover, the framework is adaptable to culturally specific scoring systems, which is essential given the growing evidence that emoji interpretation varies significantly between cultural and platform contexts. This adaptability makes the framework both future-proof and suitable for deployment in cross-cultural emotional AI studies.

6.1.4 EEC–SPS Quality Band Analysis

The dual-axis assessment based on Quality bands provides a nuanced view of how models balance emotional alignment with message fidelity.

Across all model–strategy combinations, Ideal Cases ranged from roughly one-third to just over half of outputs, with Emotion-Informed prompting yielding the highest proportions. Constrained strategies tended to produce more Semantics-Only Strong cases, suggesting that rigid preservation constraints can reduce emotional expressiveness. Conversely, Basic and Rules-Based approaches saw more Emotion-Only Strong cases, indicating that prioritizing emotional match can sometimes slightly distort the underlying text meaning.

Weak Outputs were relatively uncommon but occurred more often in lower-performing model–strategy pairs, underscoring the importance of tuning both model choice and prompting method. These patterns reinforce the study’s central finding that achieving balance between emotional and semantic quality is strategy-dependent, and that neither dimension should be optimized in isolation.

6.2 Limitations

6.2.1 Cultural and Linguistic Constraints

The most significant limitation is the exclusive focus on English text and Western emoji interpretations. Emoji meanings vary substantially across cultures, and emotional expression patterns differ between linguistic communities. Research has shown that even identical emoji characters can be interpreted differently depending on the platform rendering or the cultural background of the user (Miller, et al., 2016; Morstatter, Shu, Wang, & Liu, 2017). This variability affects both the semantic and emotional alignment scores in cross-cultural contexts.

The emotion difficulty hierarchy identified in this study may therefore reflect Western cultural norms rather than universal cognitive patterns. While the framework’s modular design allows for cultural adaptation, validating its robustness in non-Western contexts remains an essential future step.

6.2.2 Technical and Methodological Limitations

Computational Requirements: The comprehensive evaluation framework requires significant computational resources, particularly when running multi-model experiments across large datasets with multiple statistical validations. This computational demand creates potential barriers for replication, particularly in resource-constrained environments, and aligns with broader challenges in scaling multimodal AI research.

Model Version Dependency: All results are tied to specific model versions available during the data collection period. Given the rapid pace of LLM development, future iterations of the same models may show different performance patterns, making longitudinal benchmarking essential for robust conclusions.

Evaluation Framework Constraints: The EEC framework relies on automated emotion detection systems (GoEmotions, EmoTag1200), each of which has known classification limitations (Demszky et al., 2020). Any misclassifications from these systems propagate into final EEC scores, a phenomenon observed in similar evaluation pipelines for sentiment and emotion tasks (Mohammad & Bravo-Marquez, 2017). While human validation could address this, it is resource-intensive and may introduce subjective bias.

6.2.3 Scope and Generalizability Limitations

Text Type Constraints: This study is limited to short-form social media content. Longer, formal, or domain-specific texts may present challenges in emotion detection and emoji alignment that are not represented in this dataset.

Contextual Limitations: All texts were evaluated in isolation. Real-world emotional communication is context-dependent, often relying on prior conversational turns, shared history, or situational cues. This absence of conversational or multimodal context limits the ecological validity of the findings (Miller, et al., 2016).

Single-Modality Focus: The framework focuses exclusively on text-to-emoji augmentation, ignoring non-verbal modalities such as images, GIFs, and audio. These channels frequently convey emotional nuance that text alone cannot capture, and future work should explore integrated multimodal approaches (Morstatter, Shu, Wang, & Liu, 2017; Morstatter, Shu, Wang, & Liu, 2017).

6.3 Broader Impact and Contributions

6.3.1 Advancing Emotional AI Research

This dissertation provides empirical evidence about the capabilities and limitations of current LLMs in emoji augmentation tasks. It contributes a validated evaluation methodology and theoretical grounding for emotional AI research. The EEC–SPS framework offers a robust means of quantifying emotional-linguistic alignment and can be adapted across languages, cultural settings, and emotional classification schemes. Such flexibility addresses long-standing calls for standardized, reproducible evaluation in affective computing.

6.3.2 Digital Communication Evolution

Communication Technology Evolution: Emoji augmentation technologies may significantly reshape digital communication norms, much like earlier shifts caused by autocorrect and predictive text. Our findings identify quality standards and contextual considerations necessary for maintaining message authenticity in AI-augmented communication.

Accessibility and Inclusion: These systems have potential as assistive tools for individuals with communication impairments, helping them convey emotion more effectively through enriched text.

Educational Applications: The observed emotion difficulty hierarchy could inform learning platforms that train users to recognize and express emotions accurately, supporting emotional literacy initiatives.

6.3.3 Responsible AI Development

The research underscores the need for rigorous pre-deployment evaluation of emotional AI, especially in sensitive domains like healthcare and crisis communication.

Privacy and Ethical Considerations: Emotional data is highly personal; its collection, storage, and processing require privacy-preserving protocols and transparency to users (Miller et al., 2016).

Accuracy and Misrepresentation: Misaligned emoji suggestions can unintentionally distort the intended meaning of a message, potentially altering interpersonal dynamics. In contexts such as healthcare, crisis communication, or legal proceedings, this could have serious consequences. Safeguards that detect and flag low-confidence outputs or semantic-emotional mismatches should be embedded before any widespread usage.

Cultural Sensitivity: Variations in emoji interpretation underscore the importance of culturally adaptive AI systems (Morstatter, Shu, Wang, & Liu, 2017). Without such considerations, outputs may be misinterpreted across user groups, leading to communication breakdowns or perceived insensitivity.

By implementing these safeguards, developers can move toward a model of “emotionally responsible AI”, one that supports expressive richness without sacrificing accuracy, inclusivity, or ethical integrity.

6.4 Future Works

The results of this study open several high-value research directions, spanning cultural adaptation, methodological innovation, technical development, and application-specific

deployment. Given the rapid evolution of LLMs and the complex sociotechnical nature of emotional AI, these future work areas are critical for building systems that are both effective and ethically robust.

6.4.1 Immediate Research Priorities

1. *Cross-Cultural Validation Studies*: One of the most urgent next steps is to test and adapt the framework across diverse cultural and linguistic contexts. Emoji meanings and emotional interpretations vary widely between regions, languages, and communities. Future studies should therefore build culturally annotated datasets, create evaluation metrics that reflect local norms, and assess how different populations interpret emoji-emotion mappings.
2. *Longitudinal Performance Analysis*: The current evaluation captures a snapshot in time. Extending this to a longitudinal approach would reveal how model performance evolves with system updates, changes in online communication trends, or user adaptation. Such studies could also track whether LLMs maintain consistency over time or gradually drift in their emoji recommendations.
3. *Human–AI Collaborative Interfaces*: Explore interfaces where AI suggests emojis alongside an explanation of the reasoning (e.g., “added 🤔 because message contains excitement indicators”), allowing the user to accept, modify, or reject the suggestion. This aligns with calls for explainable emotional AI (XAI) in interactive systems and promotes transparency, user agency, and trust.

6.4.2 Technical Research Frontiers

1. *Multi-Modal Emotional Integration*: Emoji usage in digital communication is evolving beyond static symbols to include stickers, reactions, and other expressive elements. Future systems should integrate text-based analysis with complementary modalities such as facial expression recognition, voice-based emotion detection, and contextual image analysis. This would ensure that emoji recommendations are contextually coherent and emotionally aligned across multiple communication channels.
2. *Personalization and Adaptation*: An essential next step is to develop systems that adapt to an individual’s communication style over time. This could involve

learning user-specific preferences for emotional tone and emoji use, while incorporating privacy protections and avoiding overfitting to personal quirks.

3. *Dynamic Context-Aware Systems*: Moving beyond sentence-level analysis to incorporate conversation history, user personality traits, and situational context, enabling richer and more accurate emotional augmentation.
4. *Adaptive Learning Pipelines*: Implement continuous learning from user feedback to refine emoji mappings over time while incorporating safeguards to prevent feedback loops that reinforce bias.

6.4.3 Methodological Research Needs

1. *Human-in-the-Loop Evaluation*: Developing scalable frameworks that integrate human evaluators in iterative model refinement phases, allowing for more nuanced calibration of emotional appropriateness.
2. *Adversarial Stress Testing*: Creating adversarial test cases where textual cues conflict with typical emotional expectations (e.g., sarcasm, irony, mixed sentiment, or just neutral) to evaluate the model's robustness.
3. *Bias and Fairness Audits*: Conducting demographic analysis of emoji augmentation outputs, ensuring that suggestions do not perpetuate harmful stereotypes or systematically underperform for specific groups.
4. *Safety Protocol Development*: Establishing benchmarks and red-flag conditions for deployment in sensitive domains (e.g., detecting when an inappropriate emoji could escalate a conflict).

6.4.4 Application-Specific Research

1. *Healthcare and Therapy*: Validating emoji augmentation in therapeutic messaging platforms where nuanced emotional communication can support patient engagement and emotional well-being.
2. *Educational Technology*: Using emotion-aware emoji suggestions to teach emotional literacy, particularly for children, language learners, or individuals with autism spectrum conditions.
3. *Professional and Corporate Communication*: Investigate whether carefully moderated emoji augmentation can humanize remote workplace communication without crossing professionalism boundaries.

4. *Crisis Communication and Social Good*: Exploring emoji usage in humanitarian contexts (e.g., disaster relief, helplines) where accurate emotional signaling can improve trust and reduce misunderstandings.

The findings and recommendations presented throughout this chapter collectively address Objective O-7, translating the empirical insights from this research into actionable principles for responsible emotional AI development. By emphasizing cultural sensitivity, user-AI collaboration, transparency, and ethical safeguards, the study proposes a roadmap for deploying LLM-based emoji augmentation in ways that preserve authenticity, respect diversity, and mitigate potential harms. In doing so, it positions emotional AI not merely as a technical capability but as a socially embedded system whose design and governance must be aligned with human values and communicative integrity.

With these considerations established, the discussion now turns to the thesis conclusion. The next chapter consolidates the study's outcomes, revisits the central research question in light of the evidence gathered, and reflects on how each research objective has been fulfilled, while identifying remaining opportunities for further inquiry.

7 Conclusion

Building on the findings and recommendations outlined in Chapter 6, this chapter consolidates the outcomes of the study and explicitly maps them to the seven research objectives. The research pursued a sequence of goals spanning theoretical development, framework design, experimental validation, and applied implications.

Objectives O-1 to O-4 were addressed in Chapters 2–4 through (i) the identification of key gaps in computational affective communication, (ii) the creation of the dual-metric EEC–SPS evaluation framework, and (iii) the design and execution of a multi-model, multi-strategy experimental pipeline. O-5 was met in Chapter 5 through the analysis of performance variation across basic, secondary, and tertiary emotional dyads. O-6 was addressed in Chapter 6 by statistically examining the relationship between emotion–emoji congruence and downstream fine-grained emotion classification accuracy. Finally, O-7 was fulfilled in Chapter 6 through the synthesis of cultural, ethical, and practical guidelines for responsible emotional AI deployment.

With all research objectives met, the thesis now returns to the central research question, assessing how fully it has been answered, what conclusions can be drawn from the evidence, and where gaps remain for future investigation.

7.1 Research Question Revisited

This dissertation set out to examine the potential and limitations of LLMs in enhancing human communication using emotionally appropriate emojis. By conducting a systematic, multi-model evaluation across different prompting strategies, it was possible to provide clear, evidence-based answers to the original research questions, while also contributing to theoretical and methodological discussions within emotional AI.

Can LLMs effectively augment text with emotionally appropriate emojis while preserving semantic meaning?

Yes. Large Language Models can preserve the original meaning of text with a high degree of reliability when adding emojis. The greater challenge lies in ensuring those emojis are emotionally appropriate. While basic emotions such as joy or love are

generally handled with precision, more nuanced or blended emotions, especially dyads like *guilt* or *nostalgia*, often result in lower alignment scores. This performance gap reinforces the idea that emotional complexity remains a persistent obstacle.

The results also show notable differences between models and prompting strategies:

- Claude 3 Haiku delivered the most consistent output across diverse emotional contexts.
- GPT-3.5 Turbo achieved the highest peak score in a single scenario when paired with the strongest prompting strategy.
- Grok LLaMA3-8B performed competitively but showed greater variability depending on the emotional setting.

A key insight is that the prompting strategy consistently had more influence on performance than the underlying model architecture. In other words, *how* a model is guided often matters more than its structural design. These results also reinforce that achieving a balance between emotional congruence and semantic preservation is heavily dependent on strategy choice, and that optimizing one dimension in isolation can lead to compromises in the other. Futuristically speaking, as LLM capabilities continue to evolve, the combination of advanced architectures with well-designed prompting approaches is likely to close the current gap, achieving both high semantic preservation and accurate emotional representation.

The EEC-SPS evaluation framework developed in this research provides a dependable method for measuring these two qualities—emotional congruence and semantic preservation—separately, avoiding the risk of conflating them. It has shown strong psychometric reliability, predictive validity for related communication tasks, and the flexibility to adapt to different emotion detection systems or cultural interpretations of emotional cues. Remarkably, this framework can be used to assess new LLMs as they are developed, provided there is access to an appropriately annotated emotion dataset. Its modular structure allows the underlying tools to be updated or replaced over time, ensuring continued accuracy so long as proper mapping is maintained.

7.2 Core Contributions

This work makes three principal contributions:

- *Comprehensive empirical benchmarks:* The study provides a novel, scaled, systematic evaluation of emoji augmentation across multiple LLMs and prompting strategies, establishing performance baselines for future research. It also introduces a dual-axis quality analysis (EEC + SPS) that captures the interplay between emotional alignment and semantic preservation, allowing fine-grained comparisons across architectures, prompting strategies, and emotional categories.
- *Validated evaluation methodology:* The EEC-SPS framework delivers a robust toolset for measuring emotional and semantic dimensions of augmentation separately, offering researchers a more precise diagnostic capability.
- *Theoretical insight into emotional AI:* The research shows that emotional reasoning in current AI systems is qualitatively different from human emotional cognition. This distinction has important implications for how such systems should be designed, evaluated, and deployed in emotionally sensitive applications.

7.3 Implications and Future Directions

The results highlight that effective emotional AI systems should not aim to replace human emotional judgment but rather to complement it. In practice, this means deploying such systems in collaborative settings where the AI provides suggestions and humans make the final emotional choice.

The analysis also suggests a phased deployment strategy: initial implementations should focus on basic emotional categories, progressively incorporating more complex dyads as model capabilities improve. The modular nature of the EEC-SPS framework allows it to be adapted to alternative emotion detection methods, culturally specific emotion taxonomies, and evolving LLM capabilities. This adaptability ensures that the methodology remains relevant as emotional AI moves toward global deployment.

7.4 Final Reflection

This research highlights both the potential and the current limitations of LLM-driven emoji augmentation. It offers a balanced, evidence-based foundation for developing responsible emotional AI systems that aim to enhance rather than replace human communication.

The future of such systems lies in tools that enhance, rather than replace, human expression; tools that can adapt to cultural diversity, respect ethical boundaries, and work in partnership with users. Achieving this vision will require ongoing technical innovation, cross-cultural validation, and, above all, a commitment to responsible development.

By establishing a reliable method for evaluating emotional and semantic quality, this work sets the stage for developing emotionally intelligent AI systems that are not only technically capable but also socially aware.

8 Bibliography

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